

# Carbohydrates

# Carbohydrates

- are polyhydroxy aldehydes or ketones
- are aldehyde or ketone compounds with multiple hydroxyl groups
- $(CH_2O)_n$
- the word “saccharide” is derived from the Greek sakcharon, meaning “sugar”

# CHO Functions

- serve as *energy stores, fuels, and metabolic intermediates*.
- form part of the *structural framework of RNA and DNA*.
- are *structural elements in the cell walls of bacteria and plants*.
- are *linked to many proteins and lipids*, where they play key roles in mediating interactions among cells and interactions between cells and other elements in the cellular environment

- Diseases associated with carbohydrate metabolism include **diabetes mellitus, galactosemia, glycogen storage diseases, and lactose intolerance.**

# CHO classification

- There are major size classes of carbohydrates: monosaccharide's, disaccharide's, oligosaccharides, and polysaccharides

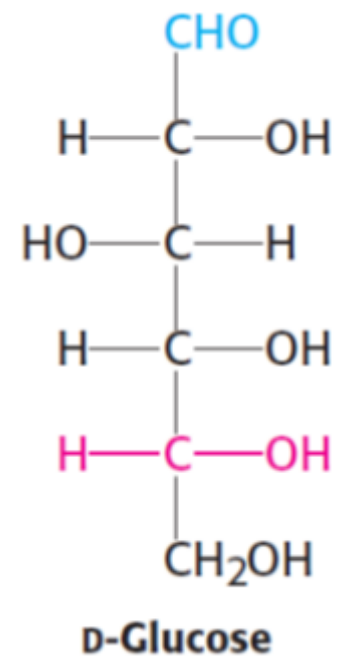
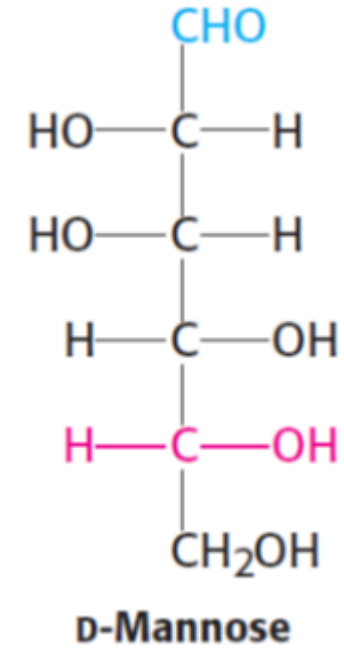
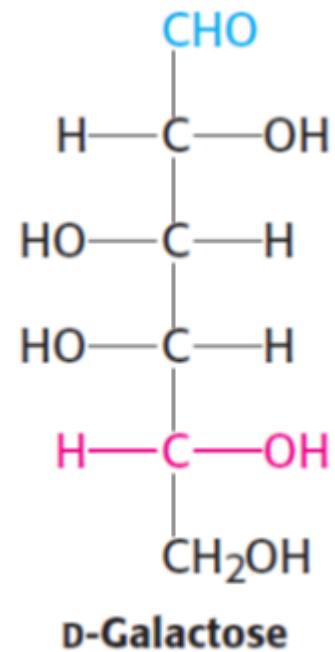
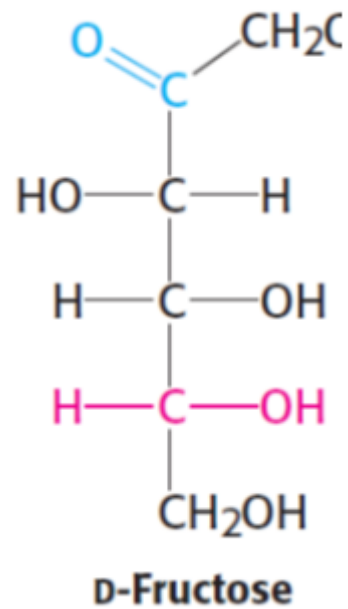
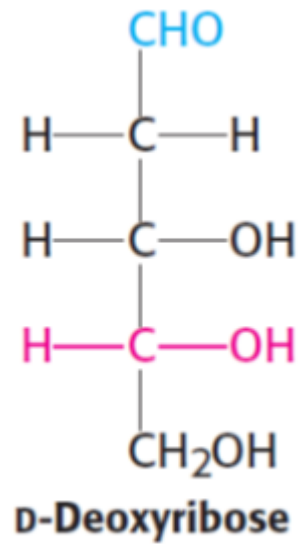
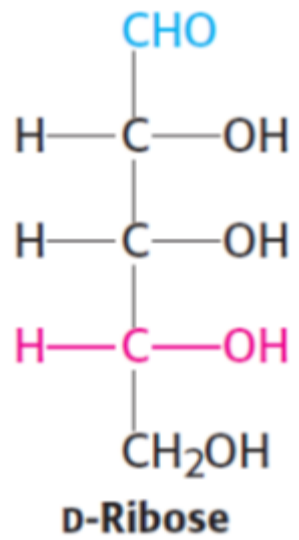
# CHO classification

- Monosaccharide's, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit.
- Oligosaccharides consist of short chains of monosaccharide units, or residues, joined by characteristic linkages called glycosidic bonds
- All common monosaccharide's and disaccharides have names ending with the suffix “-ose.”

# CHO classification

- The polysaccharides are sugar polymers containing more than 20 or so monosaccharide units, and some have hundreds or thousands of units

# Monosaccharide





# Monosaccharides

## Some Important Monosaccharides

- Glucose is the most abundant monosaccharide found in nature.
- Glucose is also known as dextrose, blood sugar.
- Glucose is broken down in cells to produce energy.

# Monosaccharides

- Galactose is found combined with glucose in the disaccharide lactose, which is present in milk and other dairy products.
- Mannose, a monosaccharide, is found in some fruits and vegetables.
- Fructose, a ketose, is commonly referred to as fruit sugar it is the sweetest monosaccharide

# Monosaccharide

- If the carbonyl group is at an end of the carbon chain (that is, in an aldehyde group) the monosaccharide is an aldose
- if the carbonyl group is at any other position (in a ketone group) the monosaccharide is a ketose.

# Monosaccharide

**TABLE 15-1 Classification of Important Sugars**

|                             | Aldoses                       | Ketoses          |
|-----------------------------|-------------------------------|------------------|
| Trioses ( $C_3H_6O_3$ )     | Glycerose<br>(glyceraldehyde) | Dihydroxyacetone |
| Tetroses ( $C_4H_8O_4$ )    | Erythrose                     | Erythrulose      |
| Pentoses ( $C_5H_{10}O_5$ ) | Ribose                        | Ribulose         |
| Hexoses ( $C_6H_{12}O_6$ )  | Glucose                       | Fructose         |
| Heptoses ( $C_7H_{14}O_7$ ) | —                             | Sedoheptulose    |

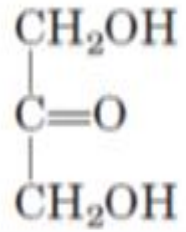
# Physiological importance of Hexoses

Hexoses of physiologic importance.

| Sugar       | Source                                                                                      | Importance                                                                                                                                                            | Clinical Significance                                                                                 |
|-------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| D-Glucose   | Fruit juices. Hydrolysis of starch, cane sugar, maltose, and lactose.                       | The "sugar" of the body. The sugar carried by the blood, and the principal one used by the tissues.                                                                   | Present in the urine (glycosuria) in diabetes mellitus owing to raised blood glucose (hyperglycemia). |
| D-Fructose  | Fruit juices. Honey. Hydrolysis of cane sugar and of inulin (from the Jerusalem artichoke). | Can be changed to glucose in the liver and so used in the body.                                                                                                       | Hereditary fructose intolerance leads to fructose accumulation and hypoglycemia.                      |
| D-Galactose | Hydrolysis of lactose.                                                                      | Can be changed to glucose in the liver and metabolized. Synthesized in the mammary gland to make the lactose of milk. A constituent of glycolipids and glycoproteins. | Failure to metabolize leads to galactosemia and cataract.                                             |
| D-Mannose   | Hydrolysis of plant mannans and gums.                                                       | A constituent of many glycoproteins.                                                                                                                                  |                                                                                                       |

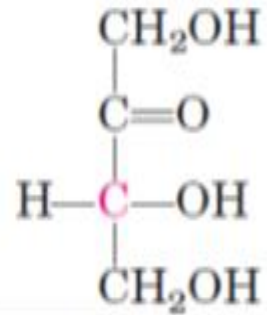


### Three carbons



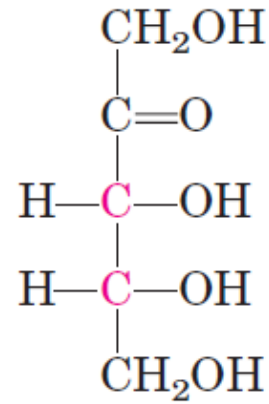
Dihydroxyacetone

### Four carbons



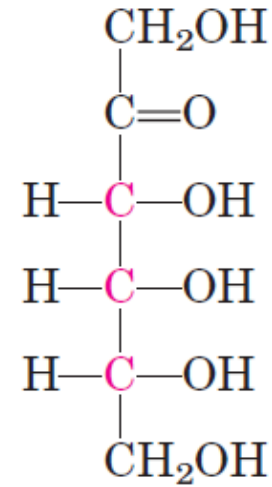
D-Erythrulose

### Five carbons

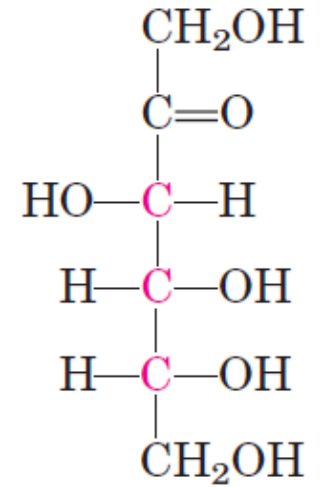


D-Ribulose

### Six carbons

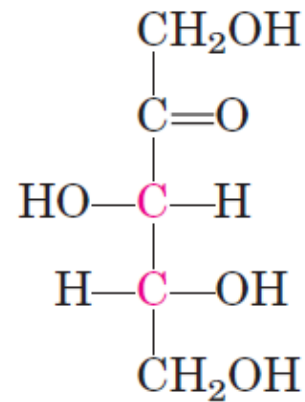


D-Psicose



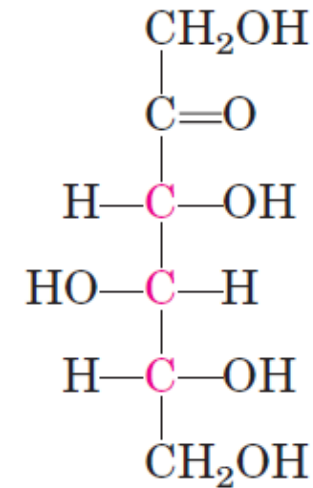
D-Fructose

## D-Ketoses (b)

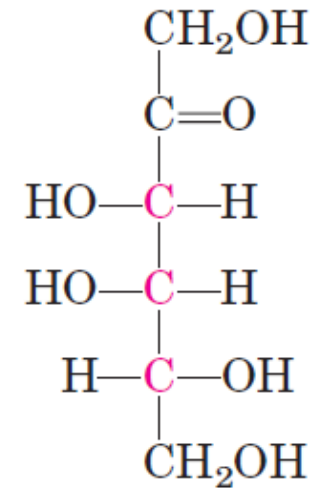


D-Xylulose

Dr. Mohamed Osama



D-Sorbose



D-Tagatose

# Structural representation of Monosaccharides

- Fisher projection: straight chain representation
- Haworth projection: simple ring in perspective
- Conformational representation: chair and boat configurations



# Isomerism

- **Isomers** are compounds with the same molecular formula but different structural formulas.
- **Asymmetric carbon (chiral)** atom is the carbon atom attached to four different atoms or chemical groups
- All carbohydrates contain at least one asymmetrical (chiral) carbon and are, therefore, optically active.

.

# Isomerism

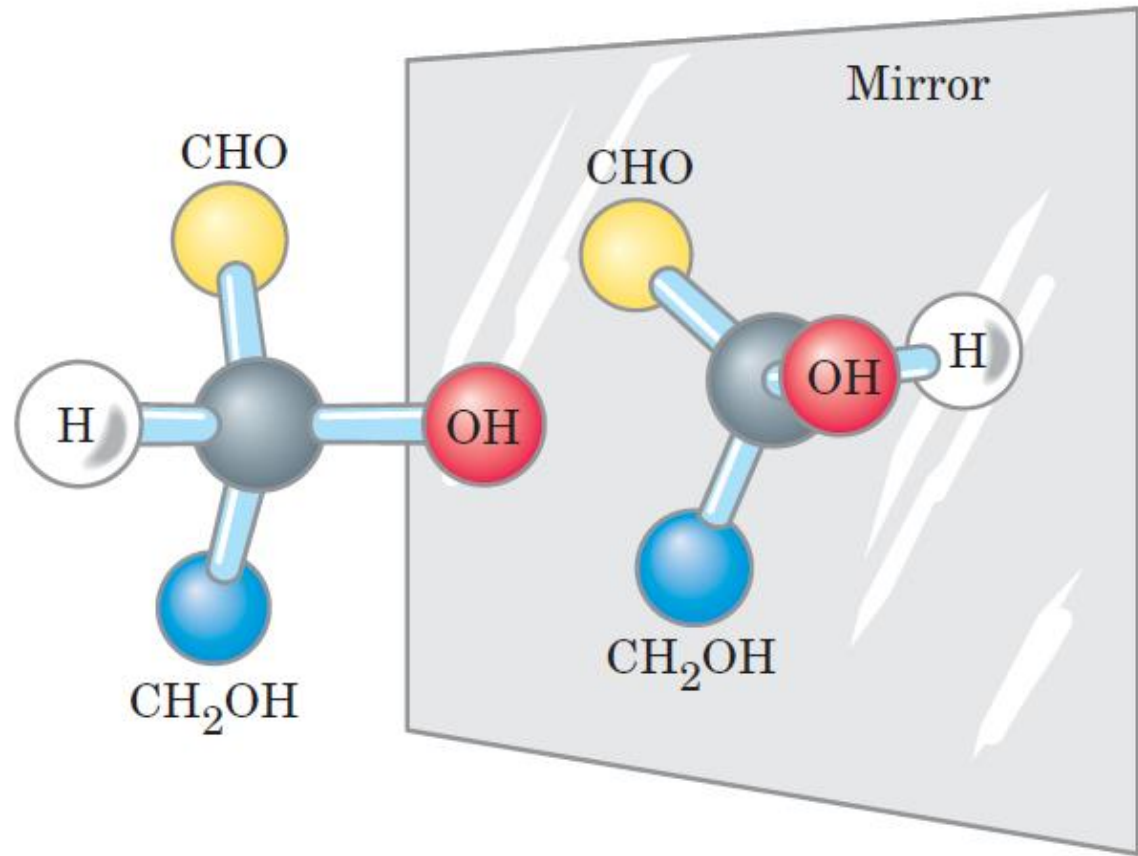
- The number of isomers of a molecule depend on the number of the chiral centers (Asymmetric carbon )in the molecule, and it follows the general formula:

$$\text{Number of isomers} = 2^n$$

n= the number of asymmetric centers (chiral centers)

- For example **glucose** contains 4 asymmetric carbon atoms So Number of isomers of glucose =  $2^4 = 16$  that means glucose has 16 isomers

# Isomerism



# Isomerism

- **The most important types of isomerism exhibited by sugars are:**

- 1- D and L isomerism

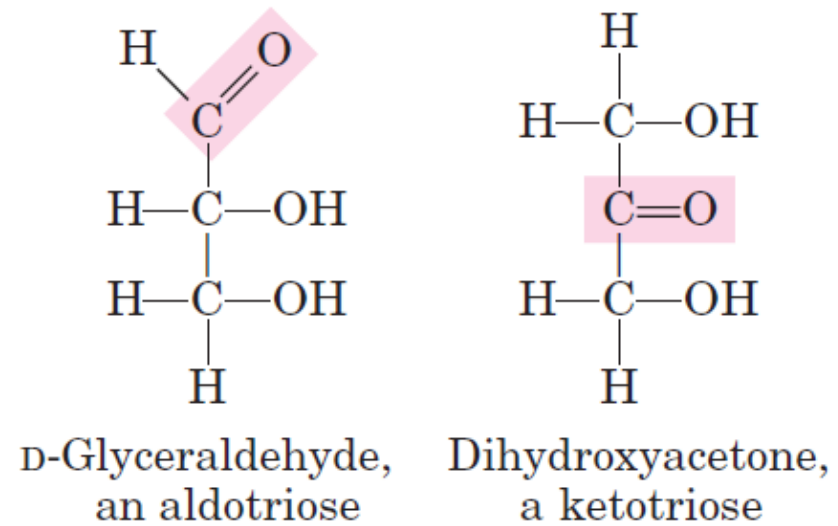
- 2- Pyranose and furanose ring structures

- 3- Alpha and beta anomers

- 4- Epimers

- 5- Aldose and ketose isomerism

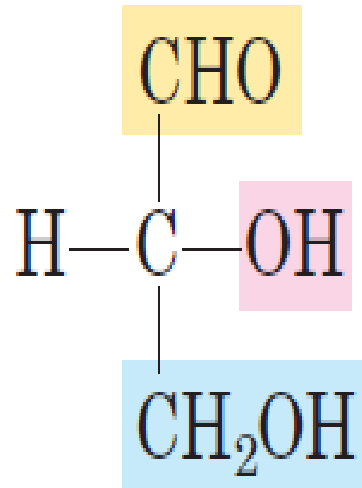
# Isomerism



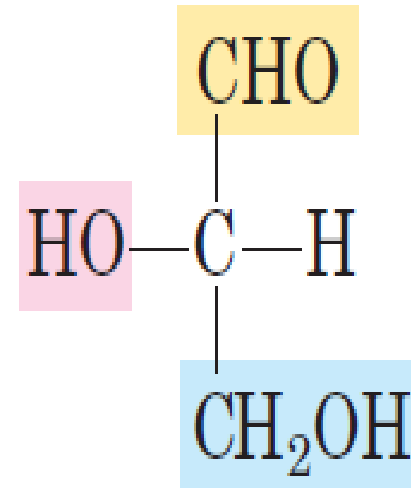
# Enantiomers

- The stereoisomers of monosaccharides of each carbon-chain length can be divided into two groups that differ in the configuration about the chiral center *most distant* from the carbonyl carbon
- When the hydroxyl group on the reference carbon is on the right in a projection formula that has the carbonyl carbon at the top, the sugar is the D isomer; when on the left, it is the L isomer

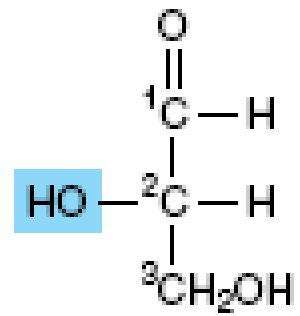
# Isomerism



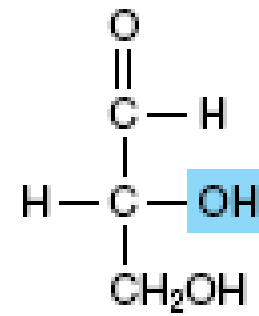
D-Glyceraldehyde



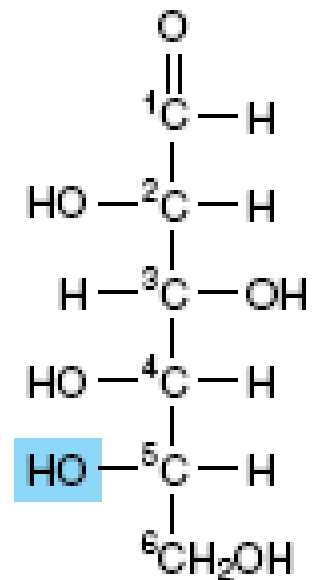
L-Glyceraldehyde



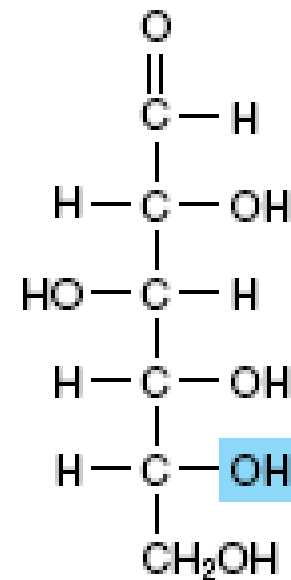
**L-Glycerose**  
(L-glyceraldehyde)



**D-Glycerose**  
(D-glyceraldehyde)

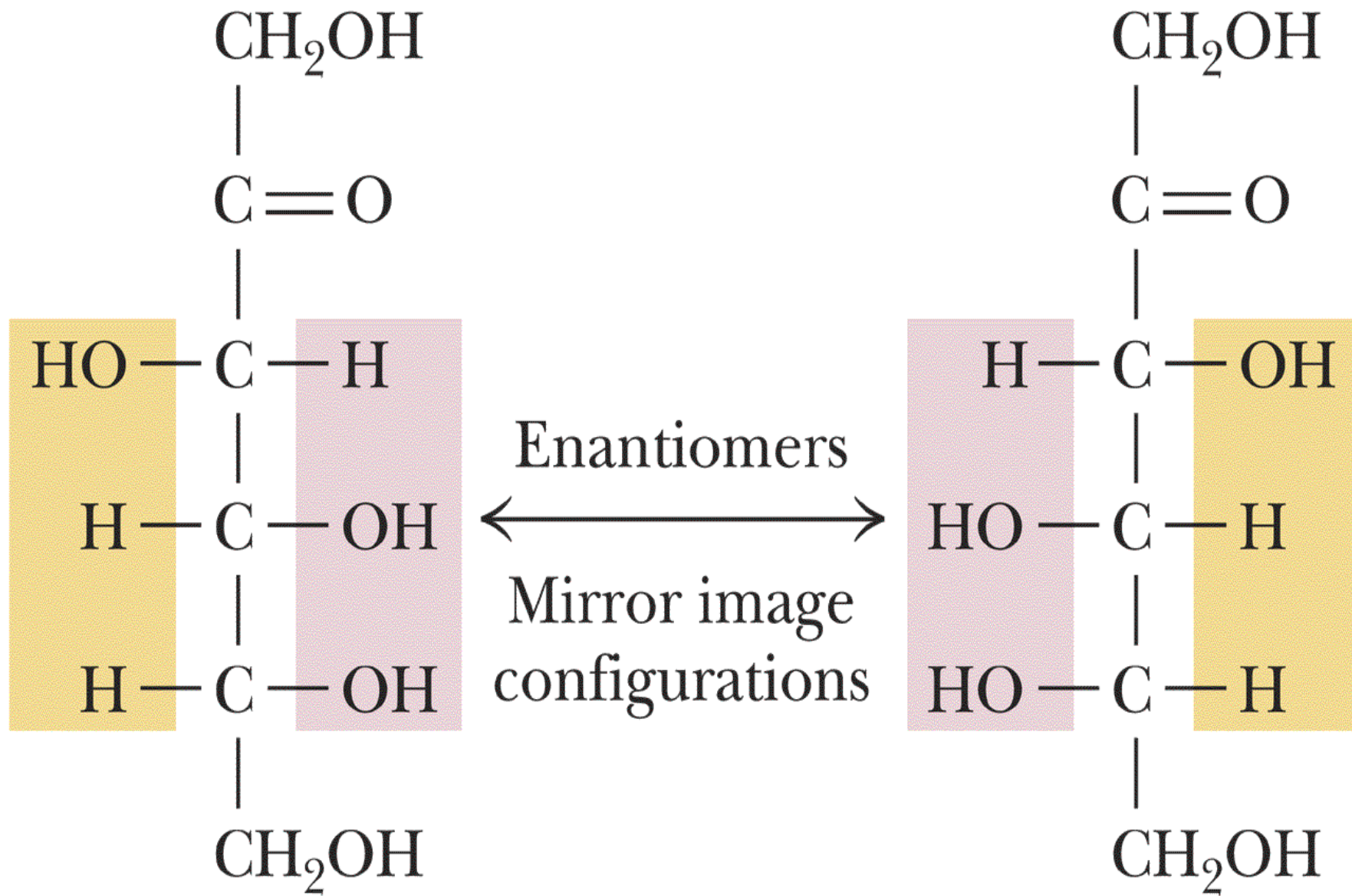


**L-Glucose**



**D-Glucose**





**D-Fructose**

**L-Fructose**

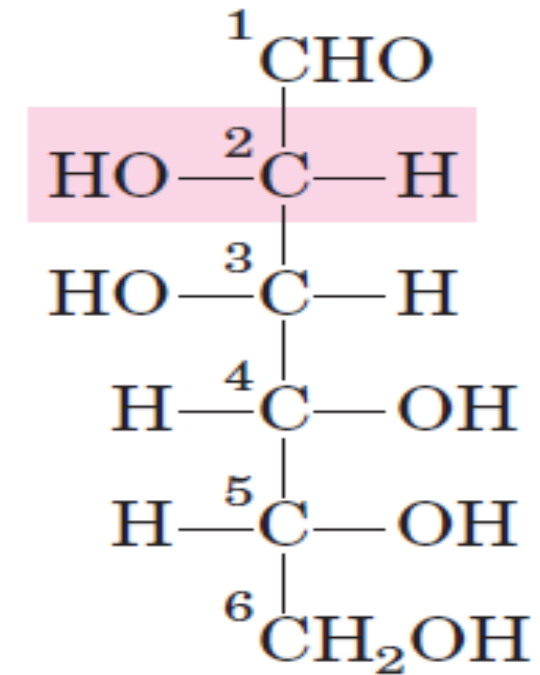
# Isomerism

- The presence of asymmetric carbon atoms also confers **optical activity** on the compound. When a beam of plane polarized light is passed through a solution of an **optical isomer**, it rotates either to the right, dextrorotatory (+), or to the left, levorotatory (–).

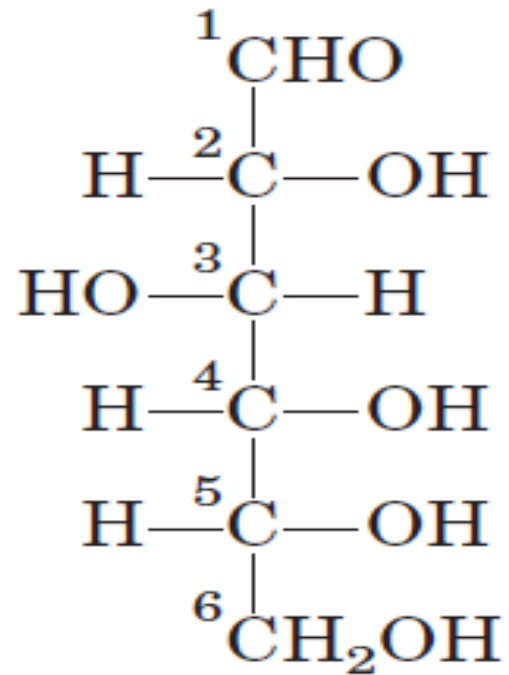
# Isomerism

- Two sugars that differ only in the configuration around one carbon atom are called **epimers**

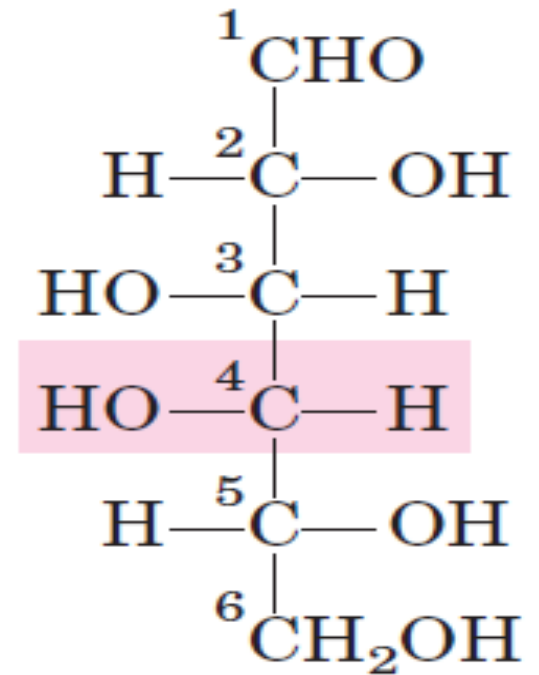
# Epimers



D-Mannose  
(epimer at C-2)



D-Glucose



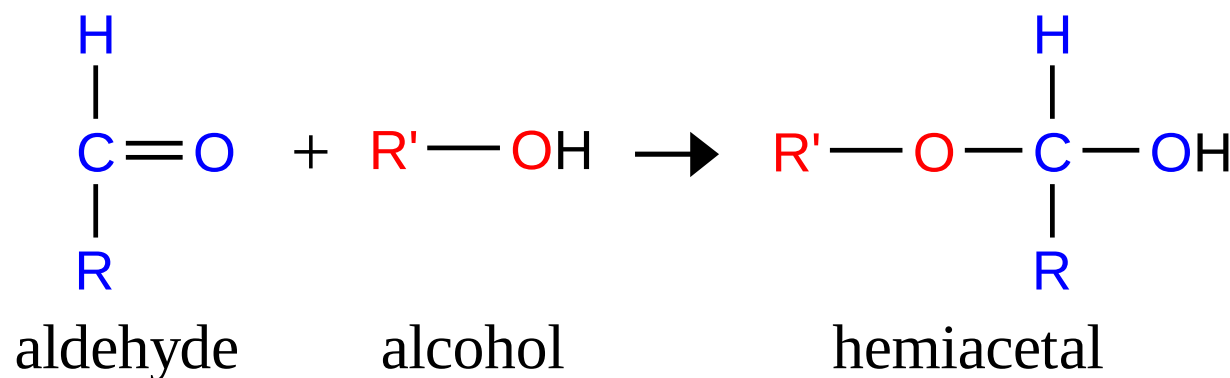
D-Galactose  
(epimer at C-4)

# Isomerism

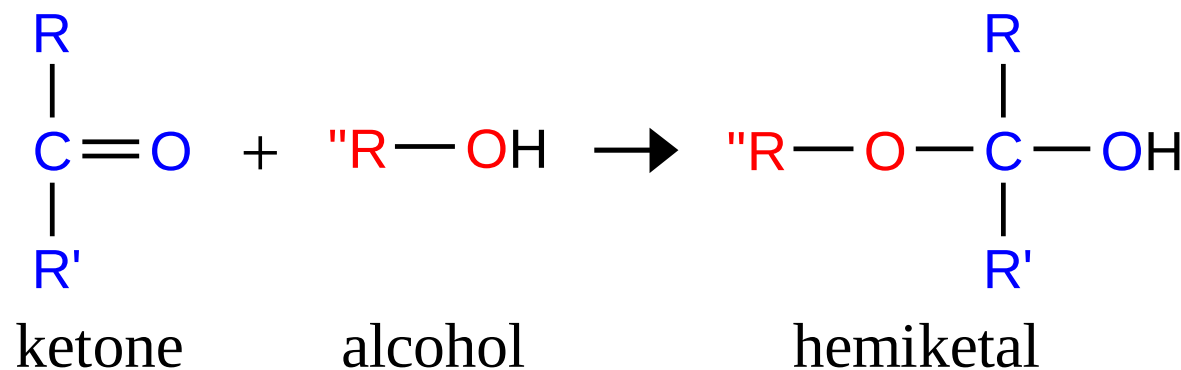
- all monosaccharaides with five or more carbon atoms in the backbone occur predominantly as cyclic (ring) structures in which the carbonyl group has formed a covalent bond with the oxygen of a hydroxyl group along the chain. The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called **hemiacetals** or **hemiketals**

# Hemiacetal & hemiketal formation

An aldehyde can react with an alcohol to form a **hemiacetal**.

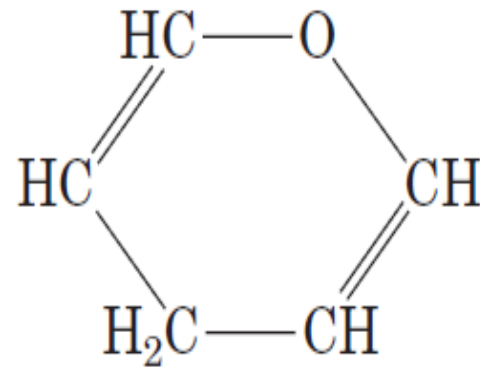


A ketone can react with an alcohol to form a **hemiketal**.

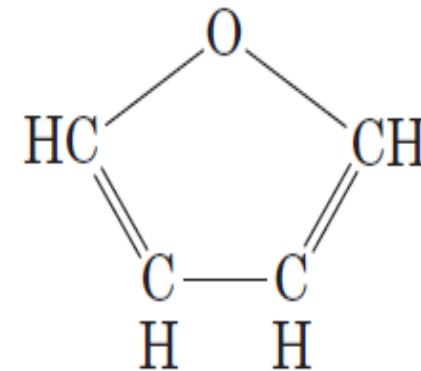


# Isomerism

- These six-membered ring compounds are called **pyranoses** because they resemble the six-membered ring compound pyran
- Aldoses also exist in cyclic forms having five membered rings, which, because they resemble the five membered ring compound furan, are called **furanoses**

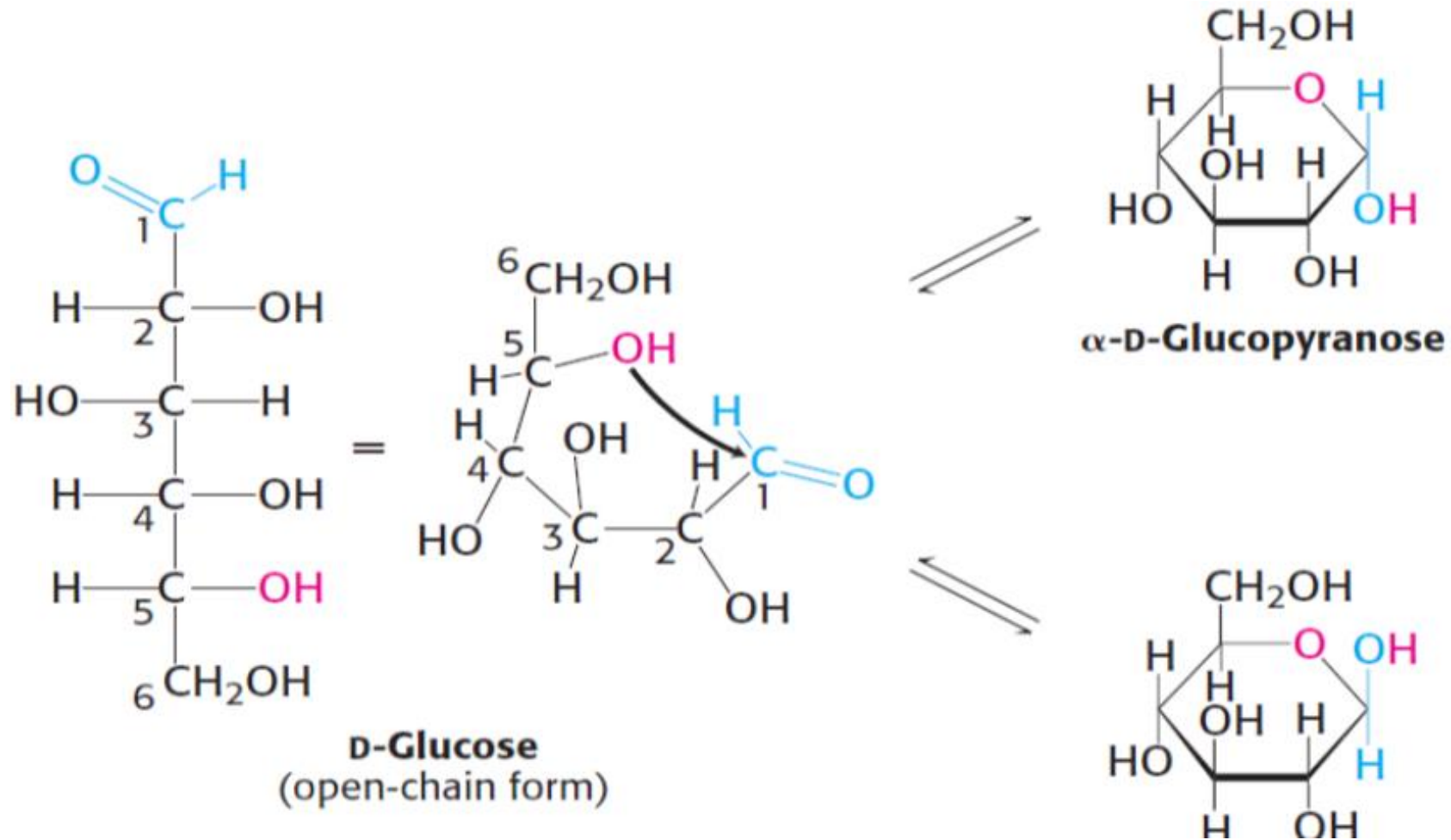


Pyran



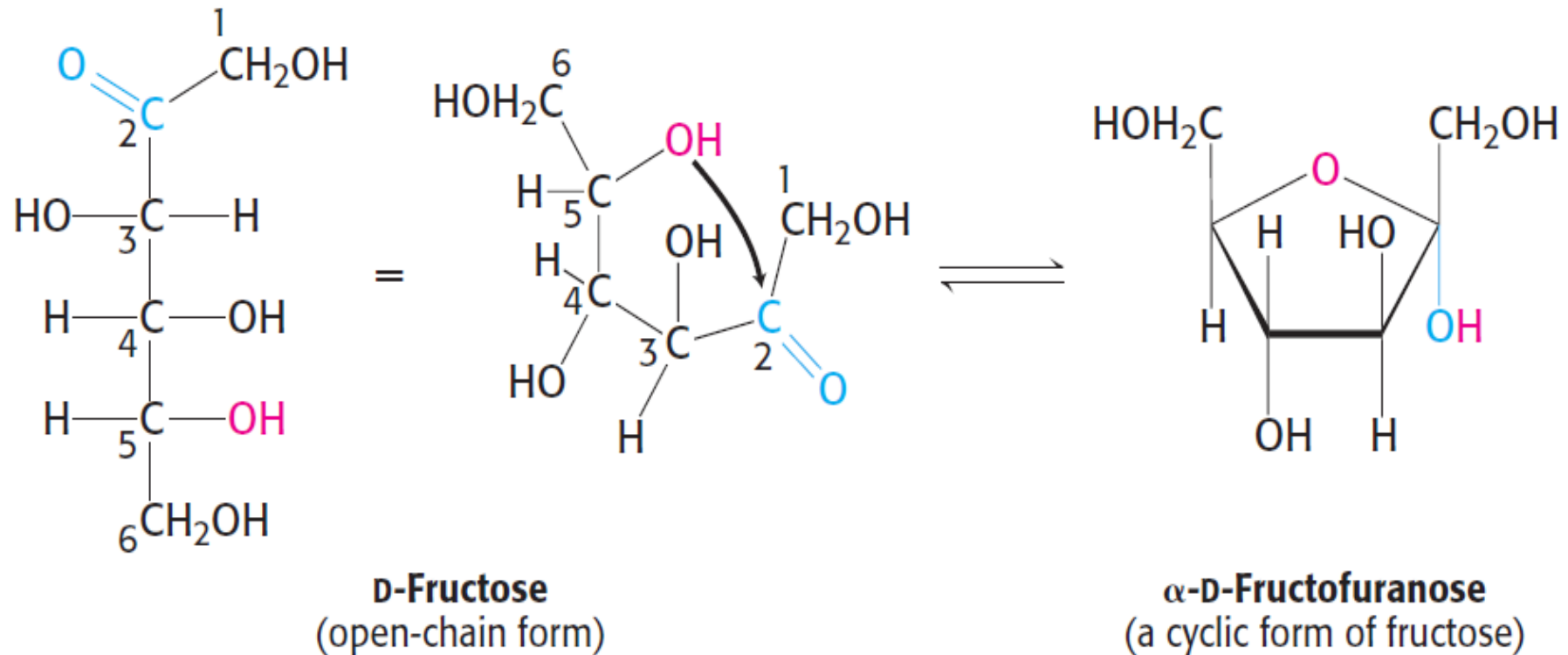
Furan

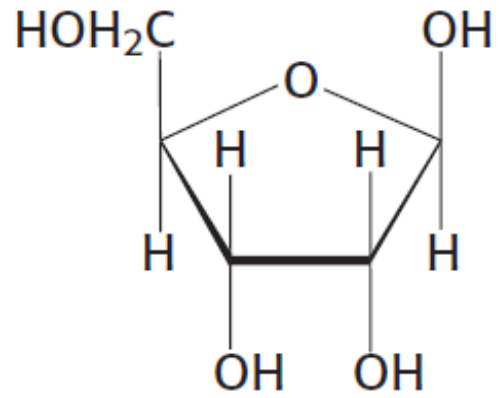
# cyclic (ring) structures



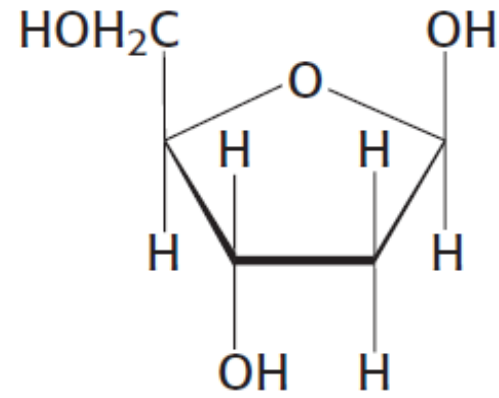


# cyclic (ring) structures

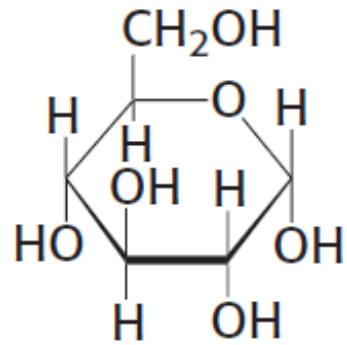




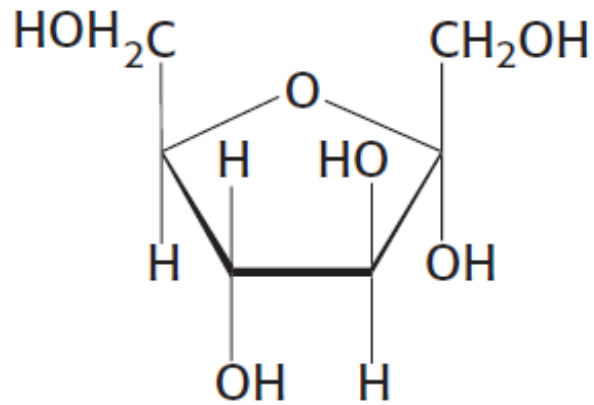
**D-Ribose**



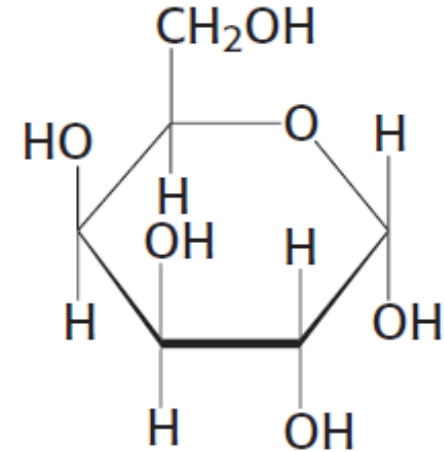
**2-Deoxy-D-ribose**



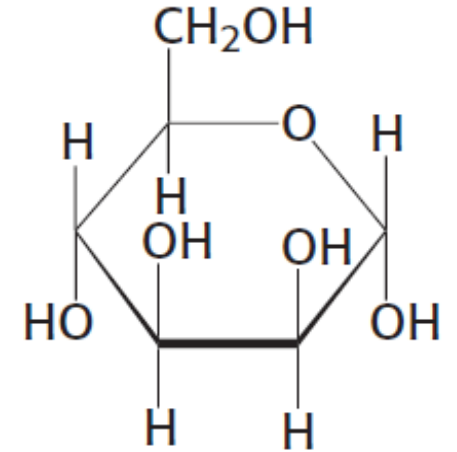
**$\alpha$ -D-Glucose**



**$\alpha$ -D-Fructose**

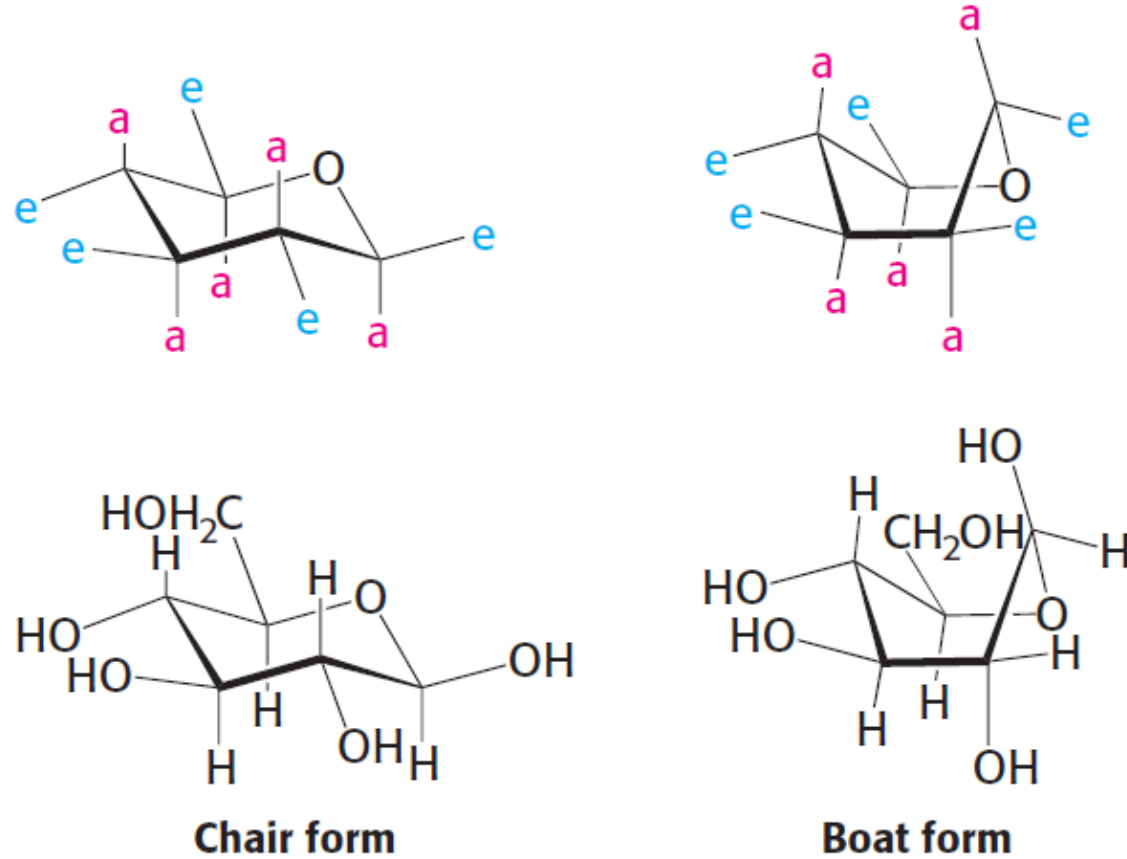


**$\alpha$ -D-Galactose**

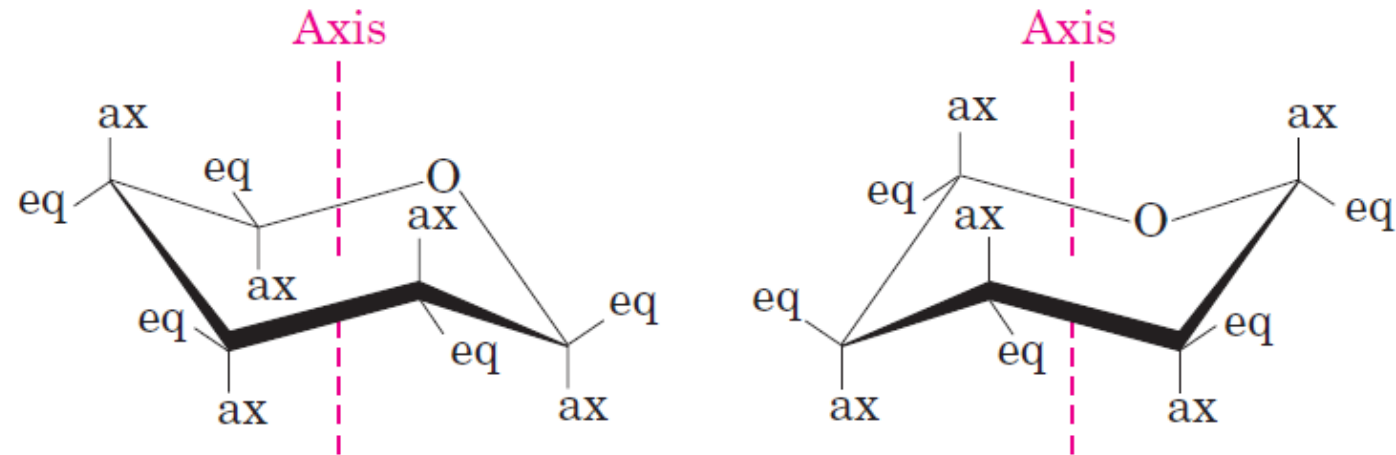


**$\alpha$ -D-Mannose**

# Isomerism



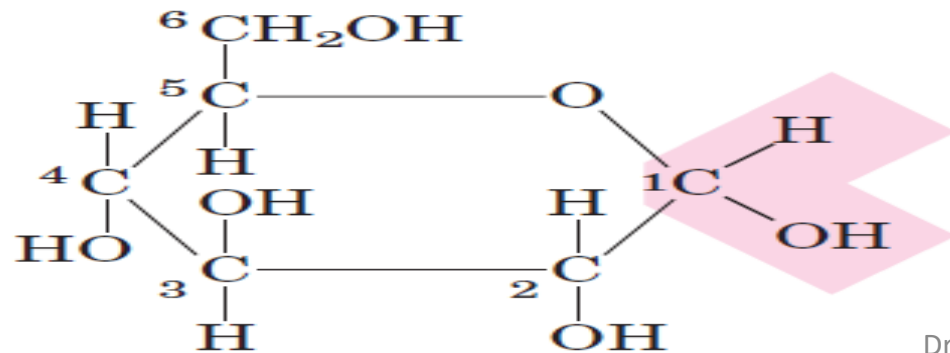
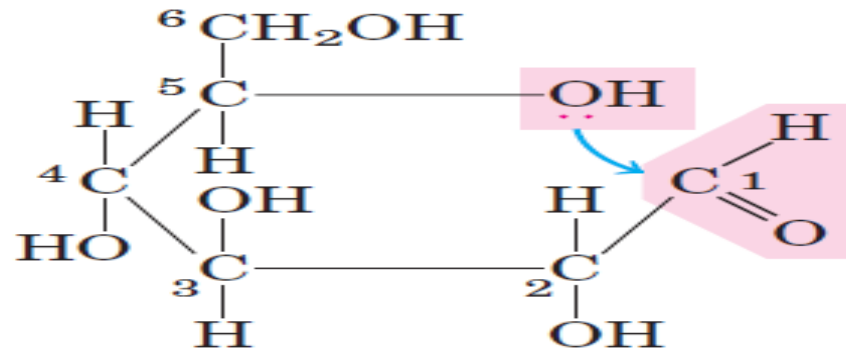
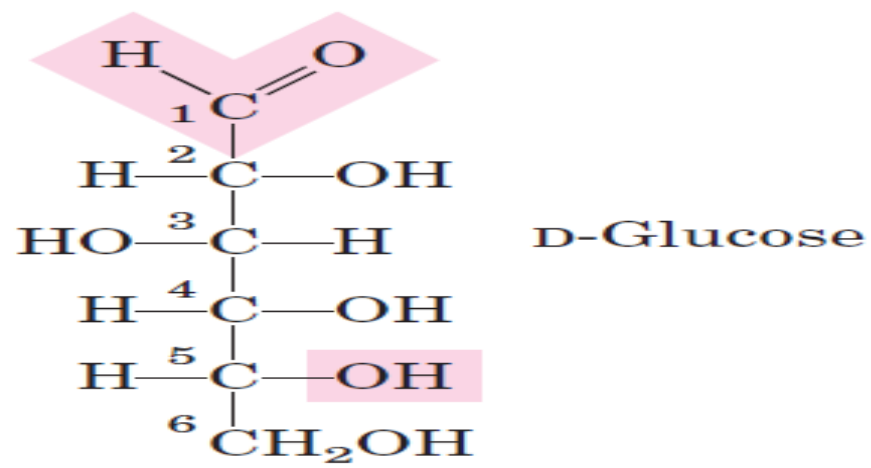
# Isomerism



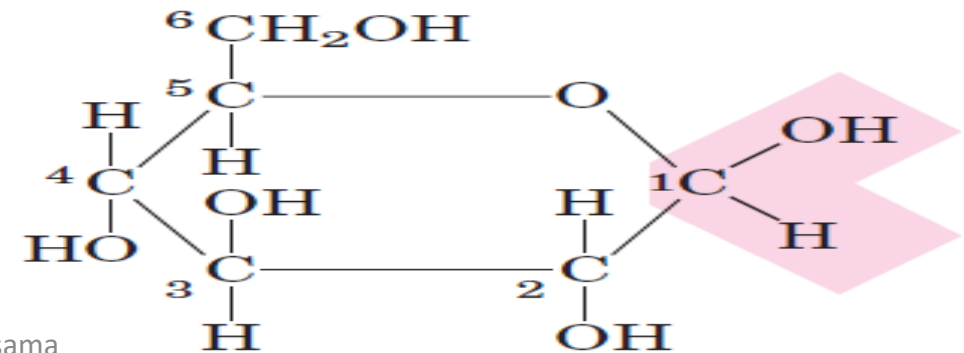
Two possible chair forms  
(a)

# Isomerism

- Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**. The hemiacetal (or carbonyl) carbon atom is called the **anomeric carbon**.
- The  $\alpha$  and  $\beta$  anomers of D-glucose interconvert in aqueous solution by a process called **mutarotation**



$\alpha$ -D-Glucopyranose



$\beta$ -D-Glucopyranose

# Isomerism

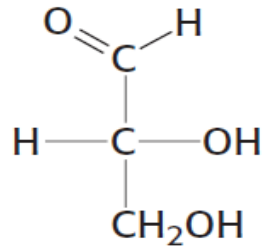
- the designation  $\alpha$  means that the hydroxyl group attached to C-1 is on the opposite side of the ring as C-6;
- $\beta$  means that the hydroxyl group is on the same side of the ring as C-6
- The C-1 carbon atom is called the *anomeric carbon atom*, and the  $\alpha$  and  $\beta$  forms are called *anomers*.

## ISOMERS

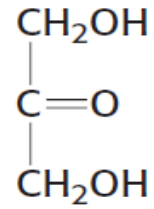
Have the same molecular formula but different structures

### CONSTITUTIONAL ISOMERS

Differ in the order of attachment of atoms



**Glyceraldehyde**  
( $\text{C}_3\text{H}_6\text{O}_3$ )



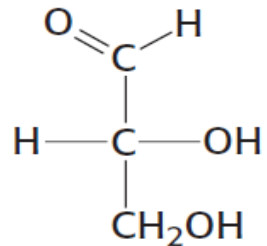
**Dihydroxyacetone**  
( $\text{C}_3\text{H}_6\text{O}_3$ )

### STEREoisomers

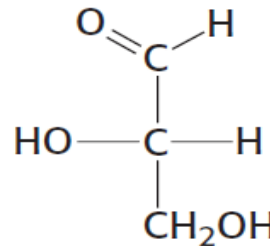
Atoms are connected in the same order but differ in spatial arrangement

### ENANTIOMERS

Nonsuperimposable mirror images



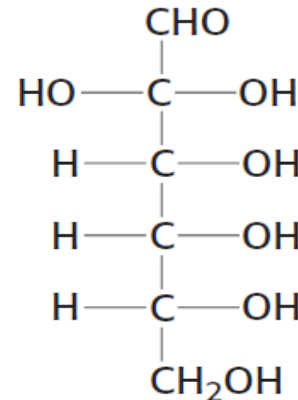
**D-Glyceraldehyde**  
( $\text{C}_3\text{H}_6\text{O}_3$ )



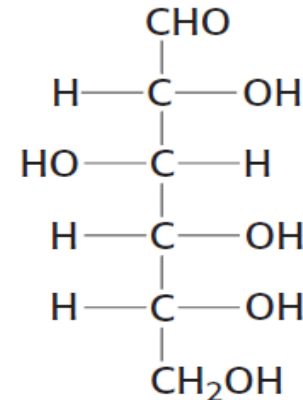
**L-Glyceraldehyde**  
( $\text{C}_3\text{H}_6\text{O}_3$ )

### DIASTEREOISOMERS

Isomers that are not mirror images



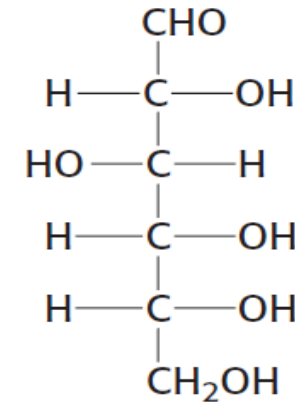
**D-Altrose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )



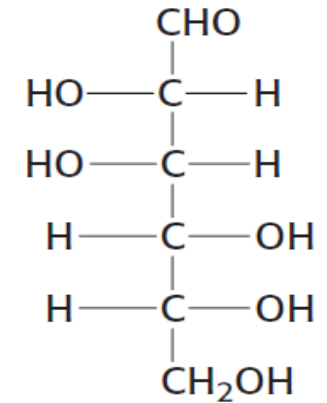
**D-Glucose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )

## EPIMERS

Differ at one of several asymmetric carbon atoms



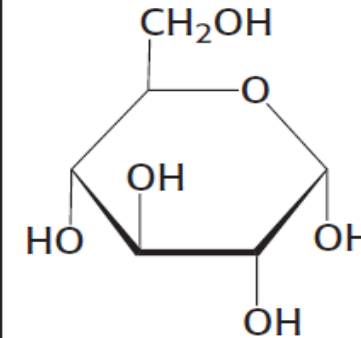
**D-Glucose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )



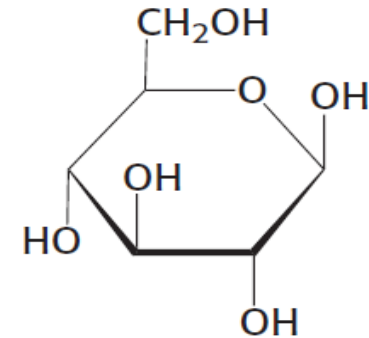
**D-Mannose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )

### ANOMERS

Isomers that differ at a new asymmetric carbon atom formed on ring closure



**$\alpha$ -D-Glucose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )



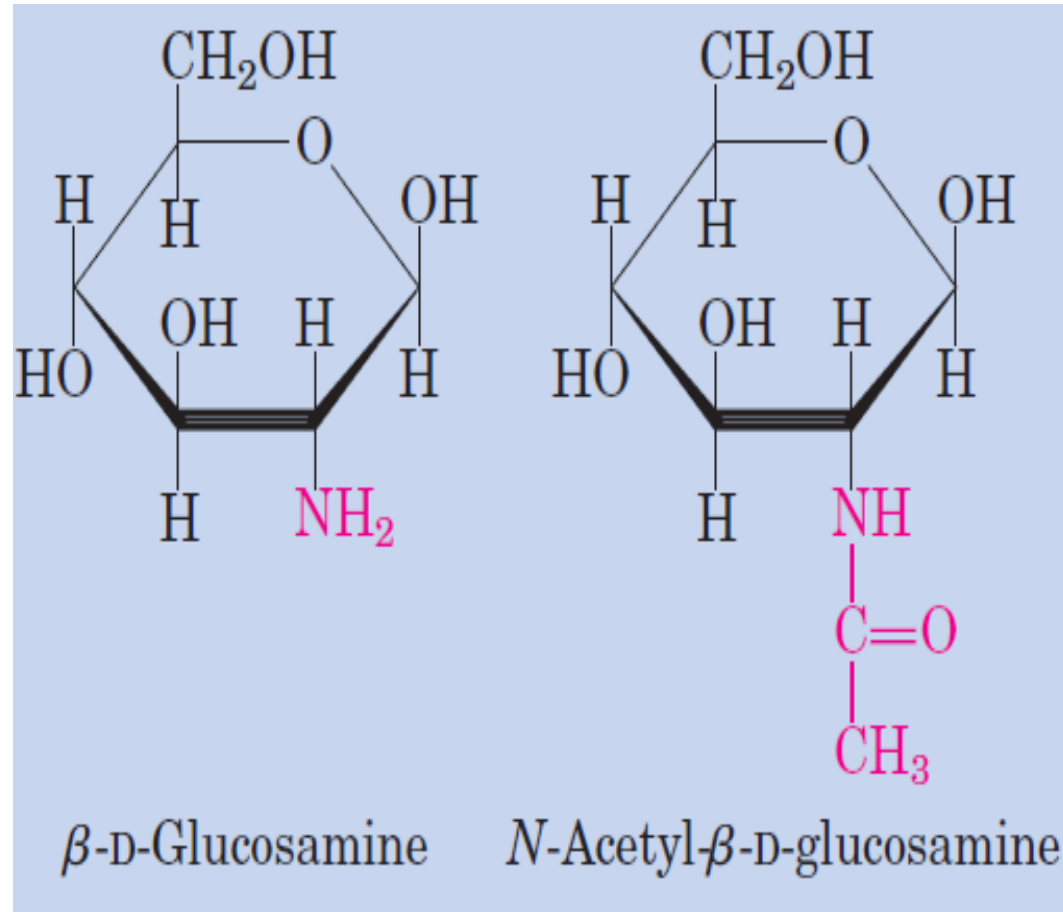
**$\beta$ -D-Glucose**  
( $\text{C}_6\text{H}_{12}\text{O}_6$ )



# Monosaccharide

- there are a number of sugar derivatives in which a hydroxyl group in the parent compound is replaced with another substituent, or a carbon atom is oxidized to a carboxyl group

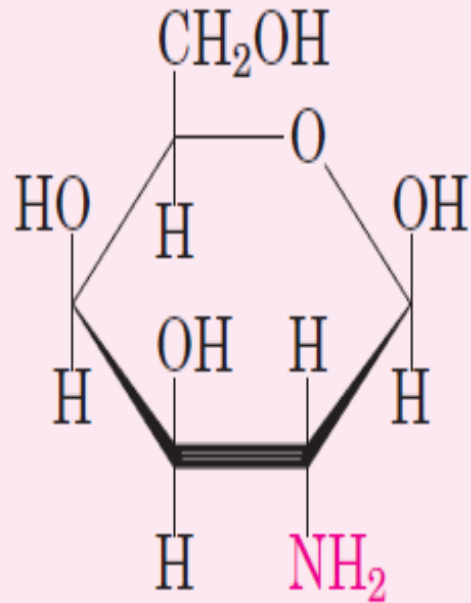
# Monosaccharide



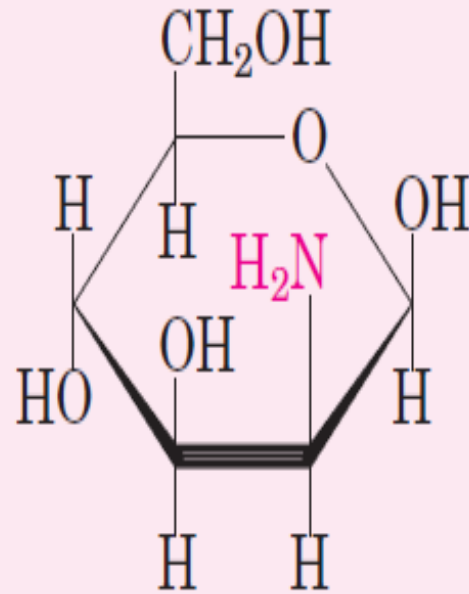
- This glucosamine derivative is part of many structural polymers, including those of the bacterial cell wall

# Monosaccharide

## Amino sugars



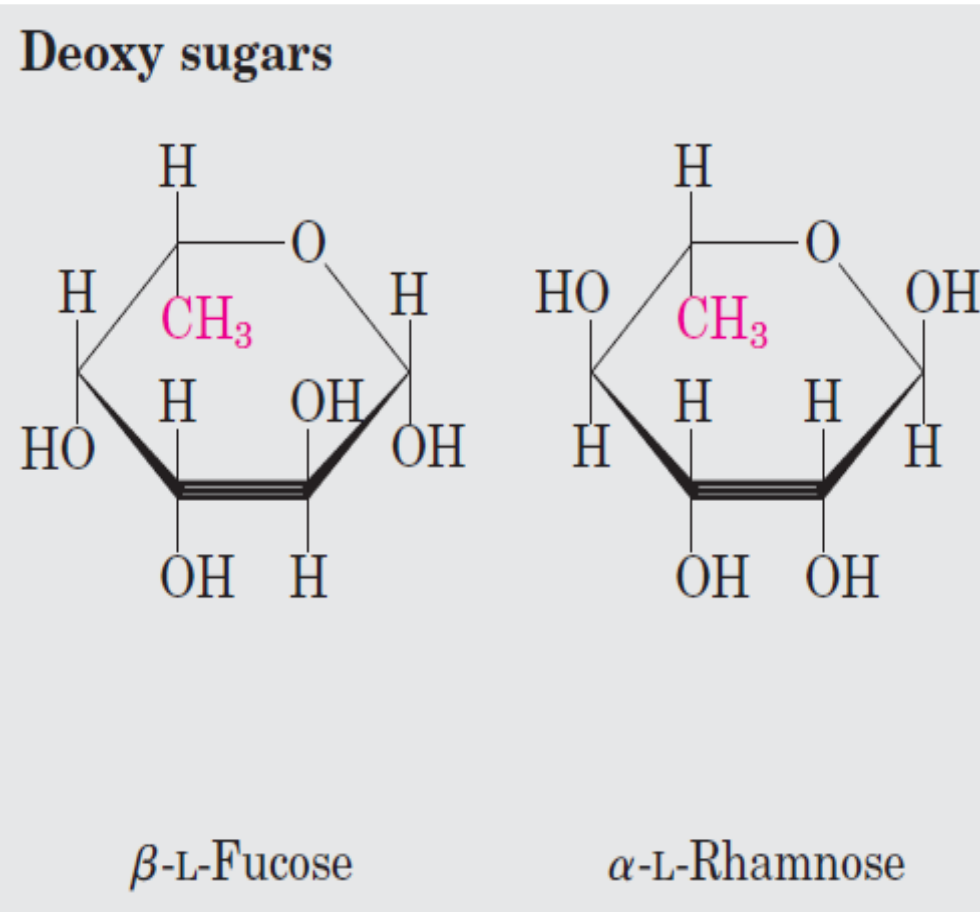
$\beta$ -D-Galactosamine



$\beta$ -D-Mannosamine

- part of many structural polymers

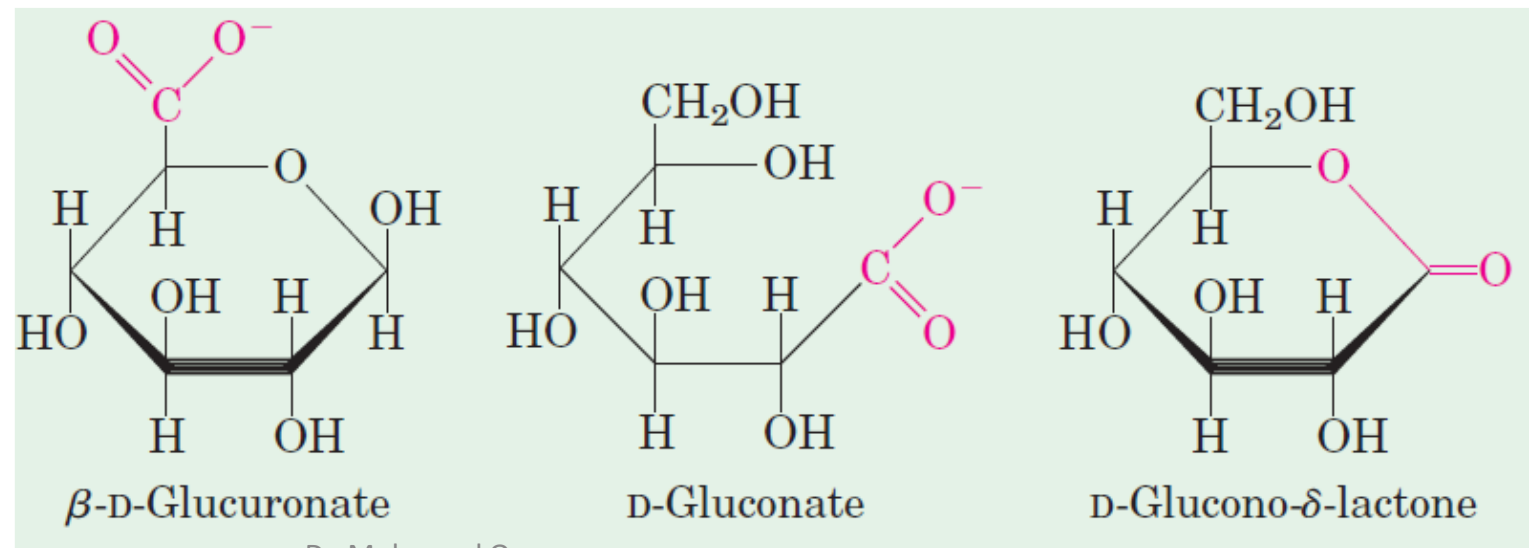
# Monosaccharide



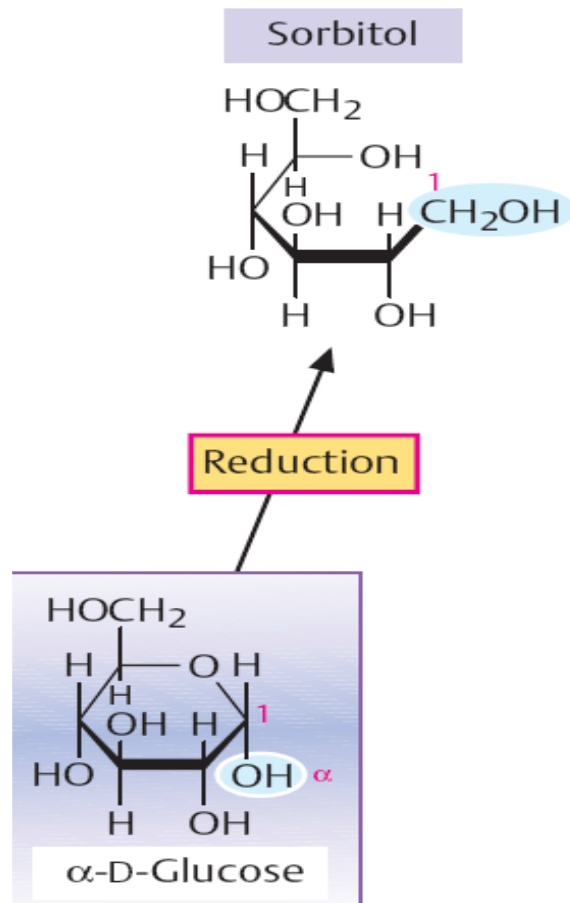
- L-Fucose is found in the complex oligosaccharide components of glycoproteins and glycolipids
- L-rhamnose is found in plant polysaccharides.

# Monosaccharide

- Oxidation of the carbonyl (aldehyde) carbon of glucose to the carboxyl level produces gluconic acid; other aldoses yield other **aldonic acids**. Oxidation of the carbon at the other end of the carbon chain—C-6 of glucose, galactose, or mannose—forms the corresponding **uronic acid**: glucuronic, galacturonic, or mannuronic acid. Both aldonic and uronic acids form stable intramolecular esters called lactones



# Monosaccharide



- sugar alcohols are formed by reduction of the carbonyl group

glucose  $\rightarrow$  glucitol (= sorbitol)

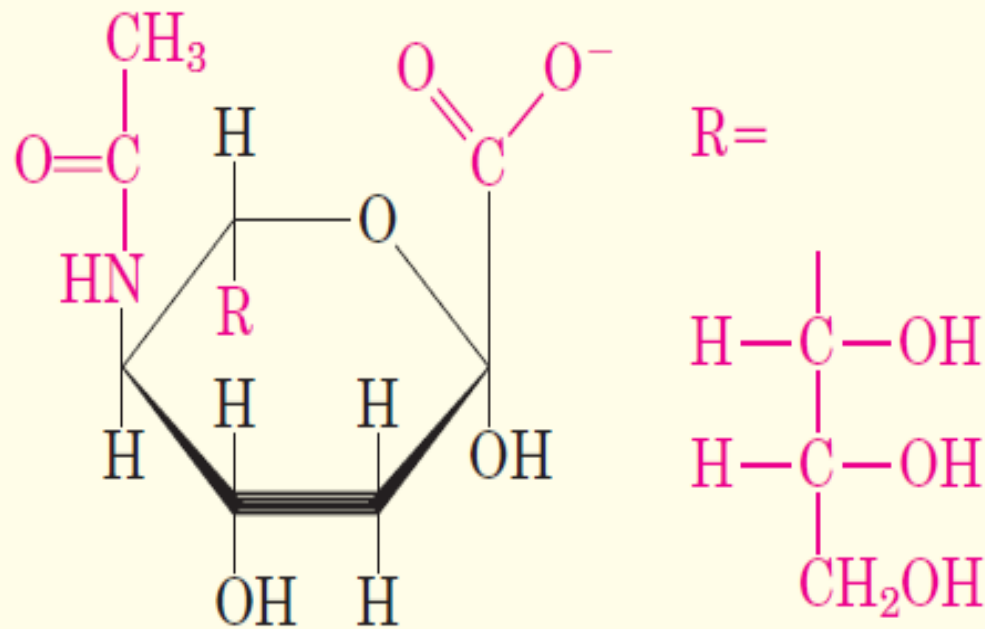
fructose  $\rightarrow$

mannose  $\rightarrow$  mannitol

galactose  $\rightarrow$  galactitol

# Monosaccharide

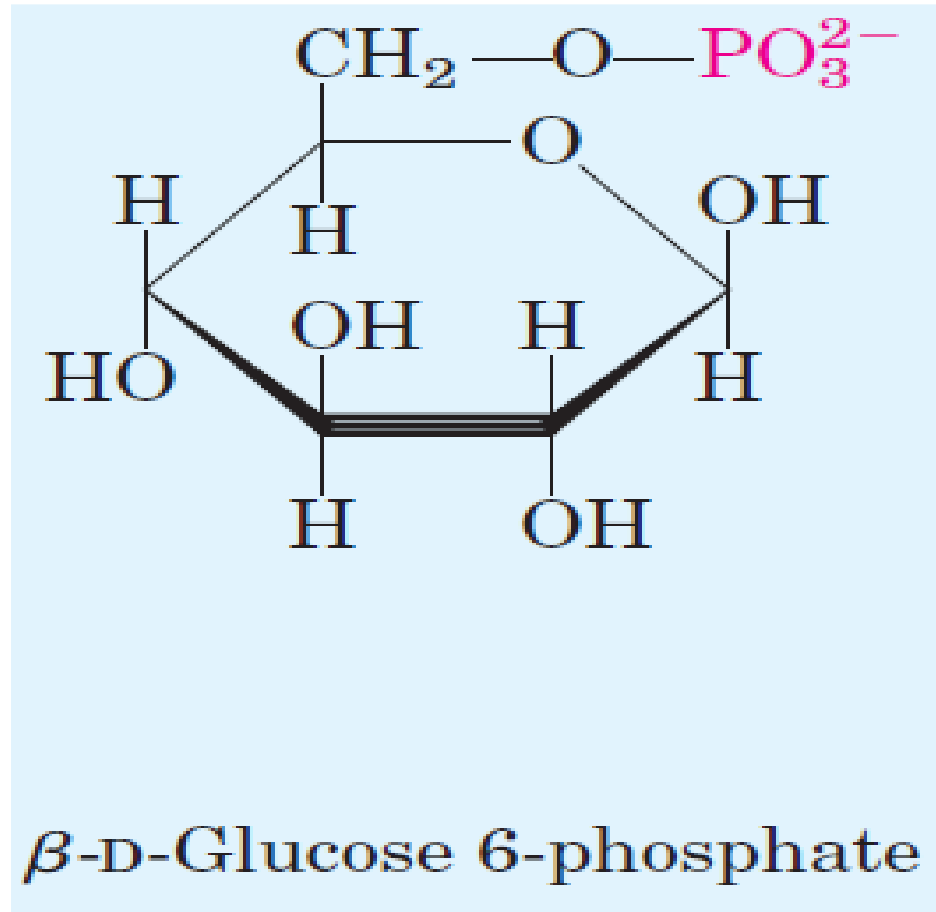
## Acidic sugars



*N*-Acetylneuraminic acid  
(a sialic acid)

- often referred to simply as “sialic acid”
- a derivative of *N*-acetylmannosamine, is a component of many glycoproteins and glycolipids in animals

# Monosaccharide

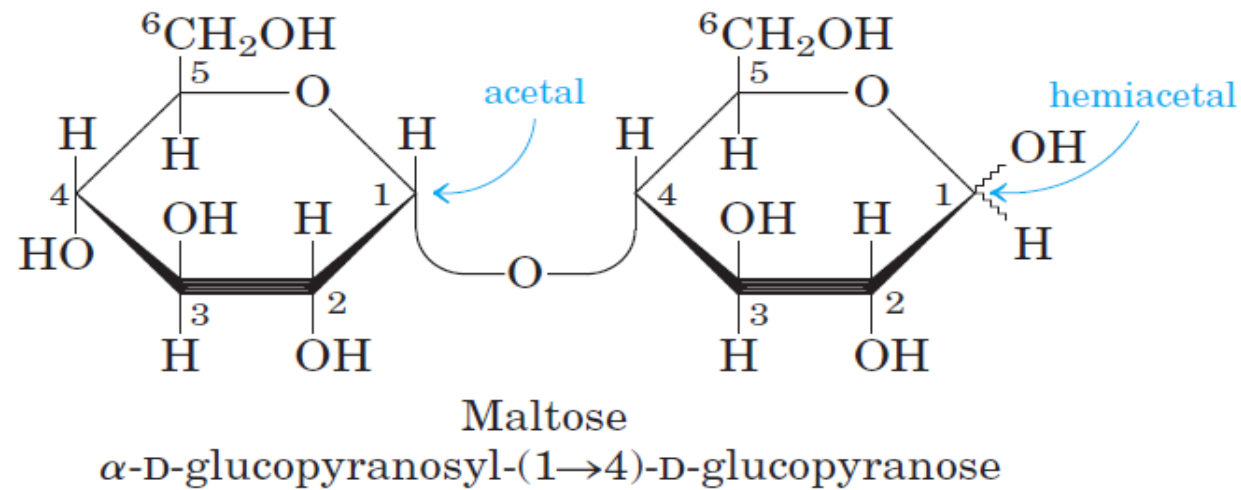
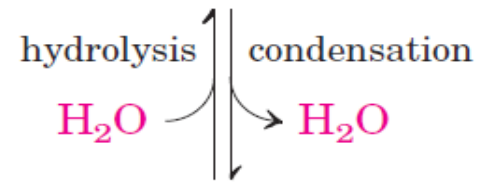
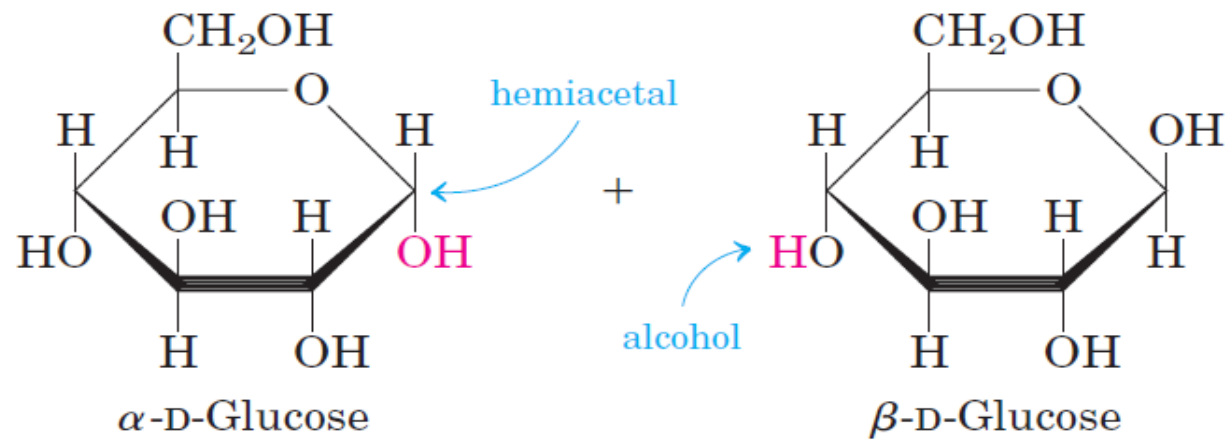


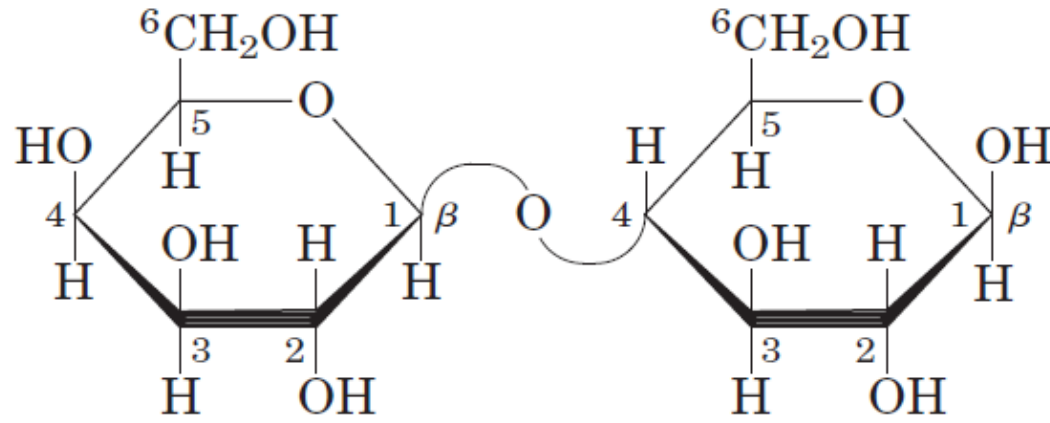
- synthesis and metabolism of carbohydrates intermediates. Sugar phosphates are relatively stable at neutral pH and bear a negative charge.
- to trap the sugar inside the cell and activates sugars for subsequent chemical transformation.
- Several important phosphorylated derivatives of sugars are components of nucleotides



# Disaccharides

- consist of two monosaccharide's joined covalently by an **O-glycosidic bond**, which is formed when a hydroxyl group of one sugar reacts with the anomeric carbon of the other
- (such as maltose, lactose, and sucrose)

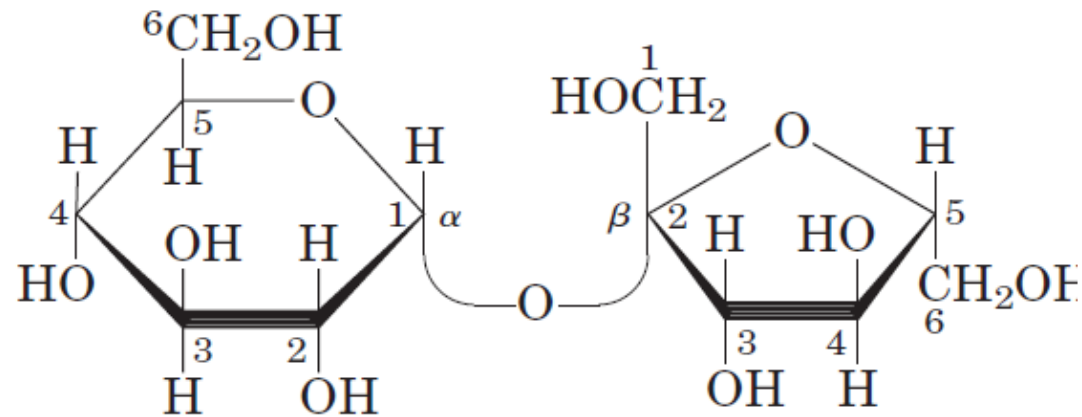




Lactose ( $\beta$  form)

$\beta$ -D-galactopyranosyl-(1 $\rightarrow$ 4)- $\beta$ -D-glucopyranose

Gal( $\beta$ 1 $\rightarrow$ 4)Glc

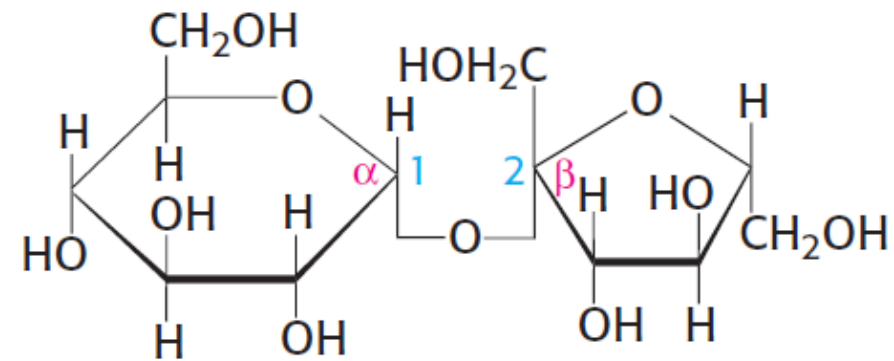


Sucrose

$\beta$ -D-fructofuranosyl  $\alpha$ -D-glucopyranoside

Fru(2 $\beta \leftrightarrow \alpha$ 1)Glc  $\equiv$  Glc( $\alpha$ 1 $\leftrightarrow$ 2 $\beta$ )Fru

# Disaccharides

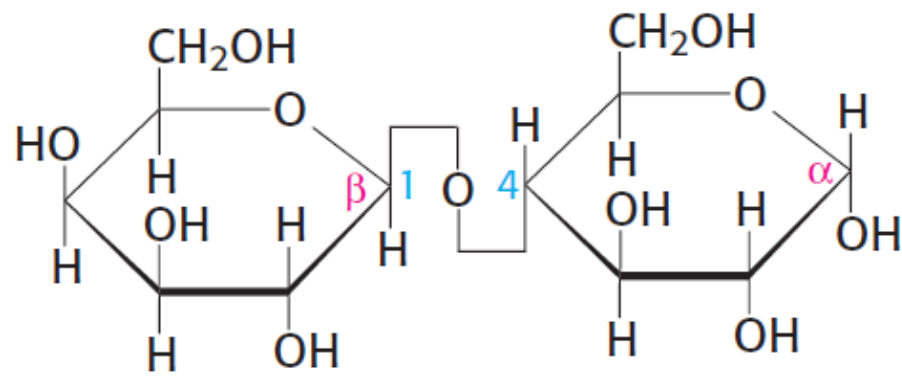


**Sucrose**  
**( $\alpha$ -D-Glucopyranosyl-(1  $\rightarrow$  2)- $\beta$ -D-fructofuranose)**

- *Sucrose* (common table sugar) is obtained commercially from sugar cane or sugar beets. The anomeric carbon atoms of a glucose unit and a fructose unit are joined in this disaccharide;
- Hydrolysis of sucrose yields a mixture of glucose and fructose called **invert sugar**

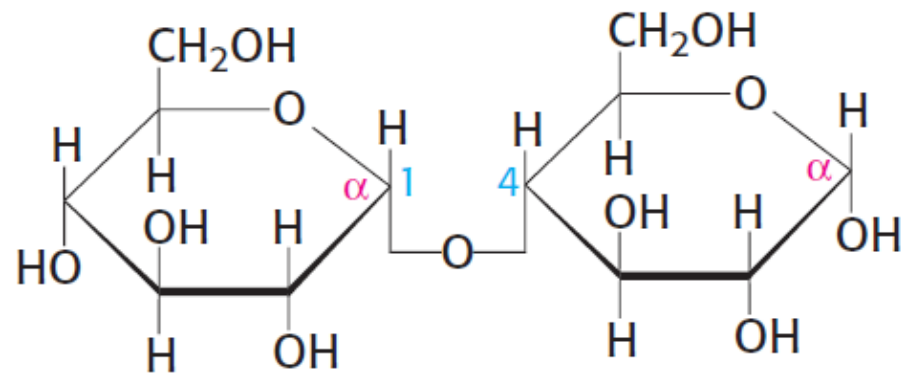
# Disaccharides

- *Lactose*, the disaccharide of milk, consists of galactose joined to glucose by a  $\beta$ -1,4-glycosidic linkage



**Lactose**  
**( $\beta$ -D-Galactopyranosyl-(1 $\rightarrow$ 4)- $\alpha$ -D-glucopyranose)**

# Disaccharides



**Maltose**  
**( $\alpha$ -D-Glucopyranosyl-(1 $\rightarrow$ 4)- $\alpha$ -D-glucopyranose)**

- *Maltose*, two glucose units are joined by an -1,4-glycosidic linkage. Maltose comes from the hydrolysis of large polymeric oligosaccharides such as starch and glycogen and is in turn hydrolyzed to glucose

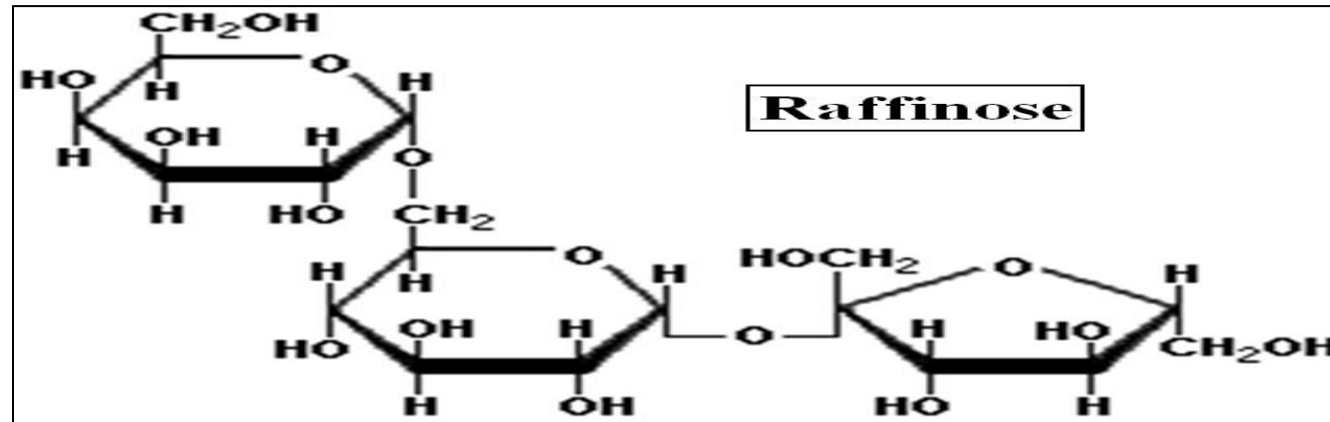
# Disaccharides

**TABLE 15–4 Disaccharides of Physiological Importance**

| Sugar      | Composition                                                                    | Source                                                               | Clinical Significance                                                                                                       |
|------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Sucrose    | O- $\alpha$ -D-glucopyranosyl-(1 $\rightarrow$ 2)- $\beta$ -D-fructofuranoside | Cane and beet sugar, sorghum and some fruits and vegetables          | Rare genetic lack of sucrase leads to sucrose intolerance—diarrhea and flatulence                                           |
| Lactose    | O- $\alpha$ -D-galactopyranosyl-(1 $\rightarrow$ 4)- $\beta$ -D-glucopyranose  | Milk (and many pharmaceutical preparations as a filler)              | Lack of lactase (alactasia) leads to lactose intolerance—diarrhea and flatulence; may be excreted in the urine in pregnancy |
| Maltose    | O- $\alpha$ -D-glucopyranosyl-(1 $\rightarrow$ 4)- $\alpha$ -D-glucopyranose   | Enzymic hydrolysis of starch (amylase); germinating cereals and malt |                                                                                                                             |
| Isomaltose | O- $\alpha$ -D-glucopyranosyl-(1 $\rightarrow$ 6)- $\alpha$ -D-glucopyranose   | Enzymic hydrolysis of starch (the branch points in amylopectin)      |                                                                                                                             |
| Lactulose  | O- $\alpha$ -D-galactopyranosyl-(1 $\rightarrow$ 4)- $\beta$ -D-fructofuranose | Heated milk (small amounts), mainly synthetic                        | Not hydrolyzed by intestinal enzymes, but fermented by intestinal bacteria; used as a mild osmotic laxative                 |
| Trehalose  | O- $\alpha$ -D-glucopyranosyl-(1 $\rightarrow$ 1)- $\alpha$ -D-glucopyranoside | Yeasts and fungi; the main sugar of insect hemolymph                 |                                                                                                                             |

# Oligosaccharides

- Oligosaccharides are condensation product of three to ten monosaccharides
- Most are not digested by human enzymes
- Raffinose, also called melitose, is a trisaccharides





# Polysaccharides

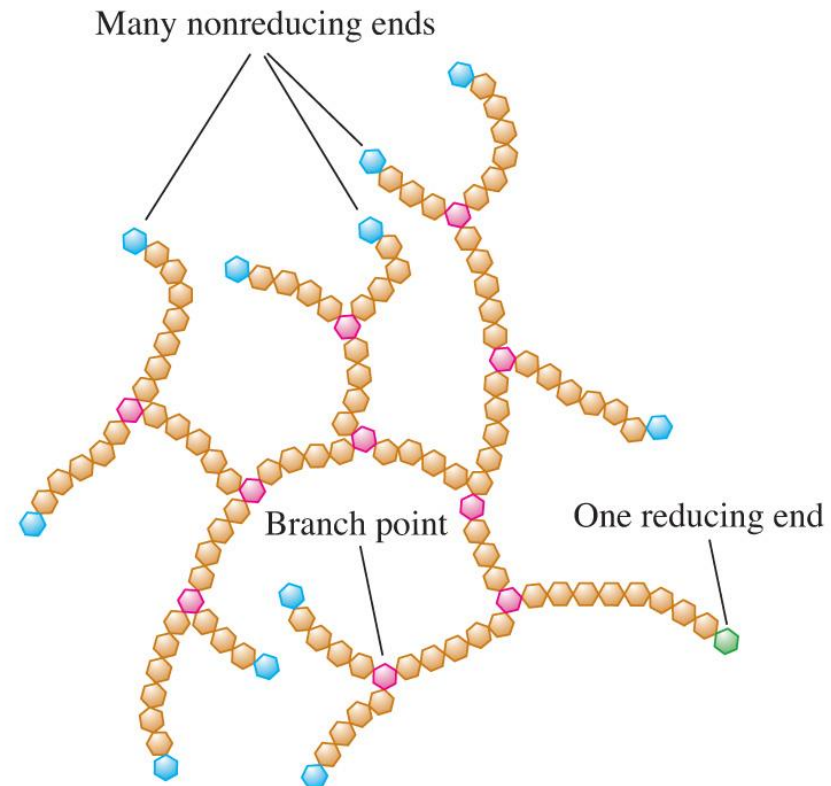
- Most carbohydrates found in nature occur as polysaccharides, polymers of medium to high molecular weight.
- Polysaccharides, also called **glycans**, differ from each other in the identity of their recurring monosaccharide units, in the length of their chains, in the types of bonds linking the units, and in the degree of branching.

# Structure of Polysaccharides

a.



b.

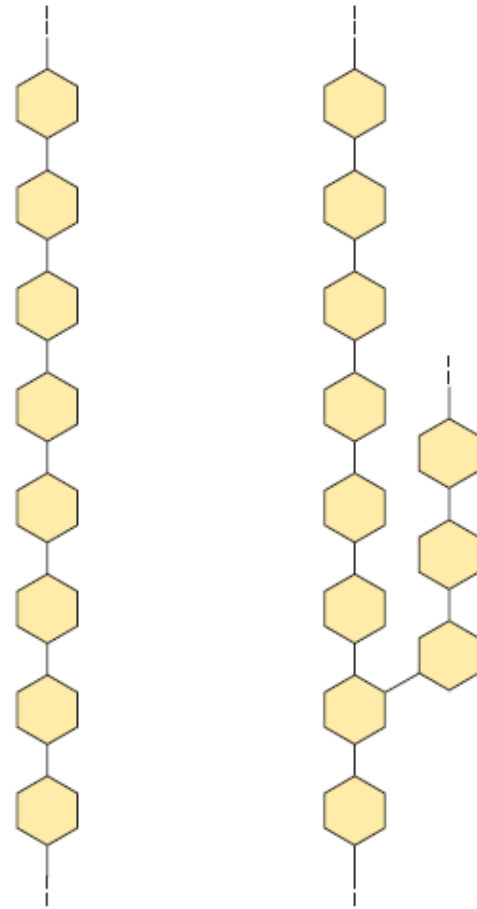


# Polysaccharides

- **Homopolysaccharides** contain only a single monomeric species;
- **Heteropolysaccharides** contain two or more different kinds

## Homopolysaccharides

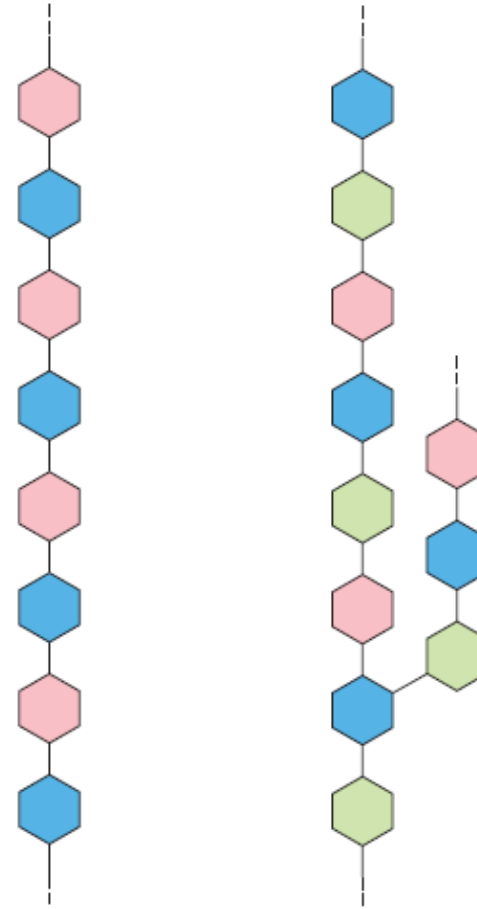
Unbranched    Branched



## Heteropolysaccharides

Two  
monomer  
types,  
unbranched

Multiple  
monomer  
types,  
branched



# Polysaccharides

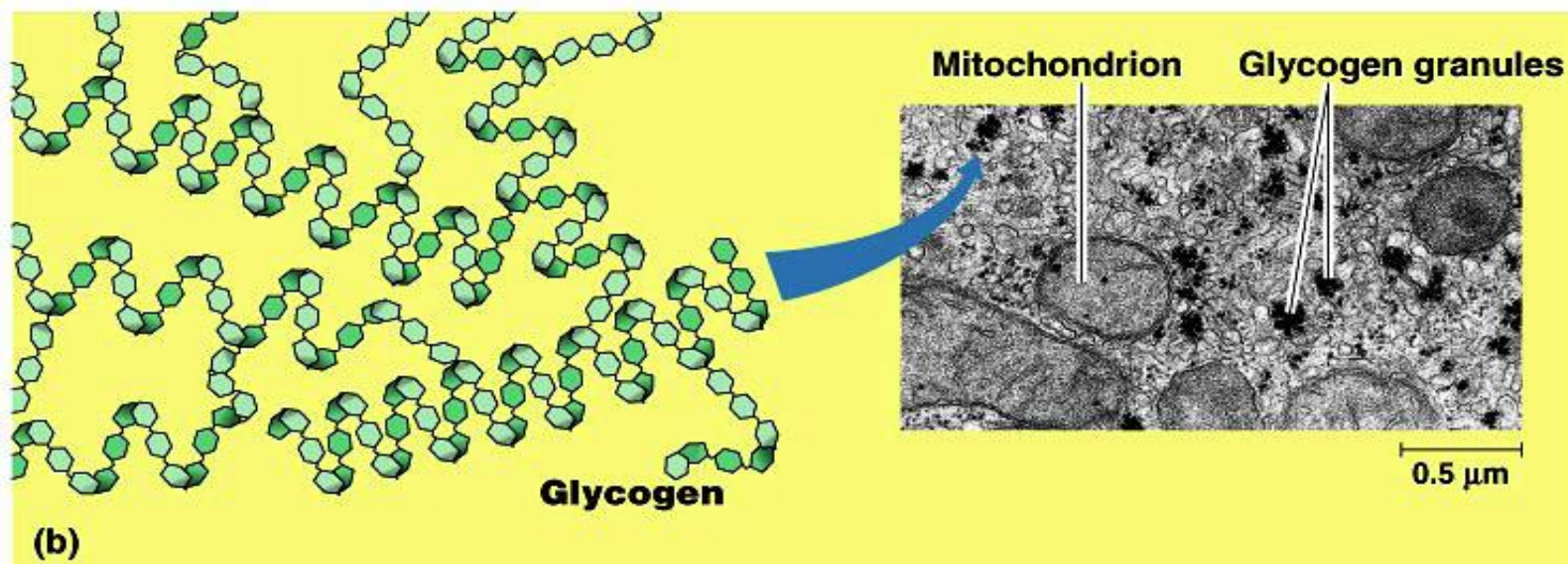
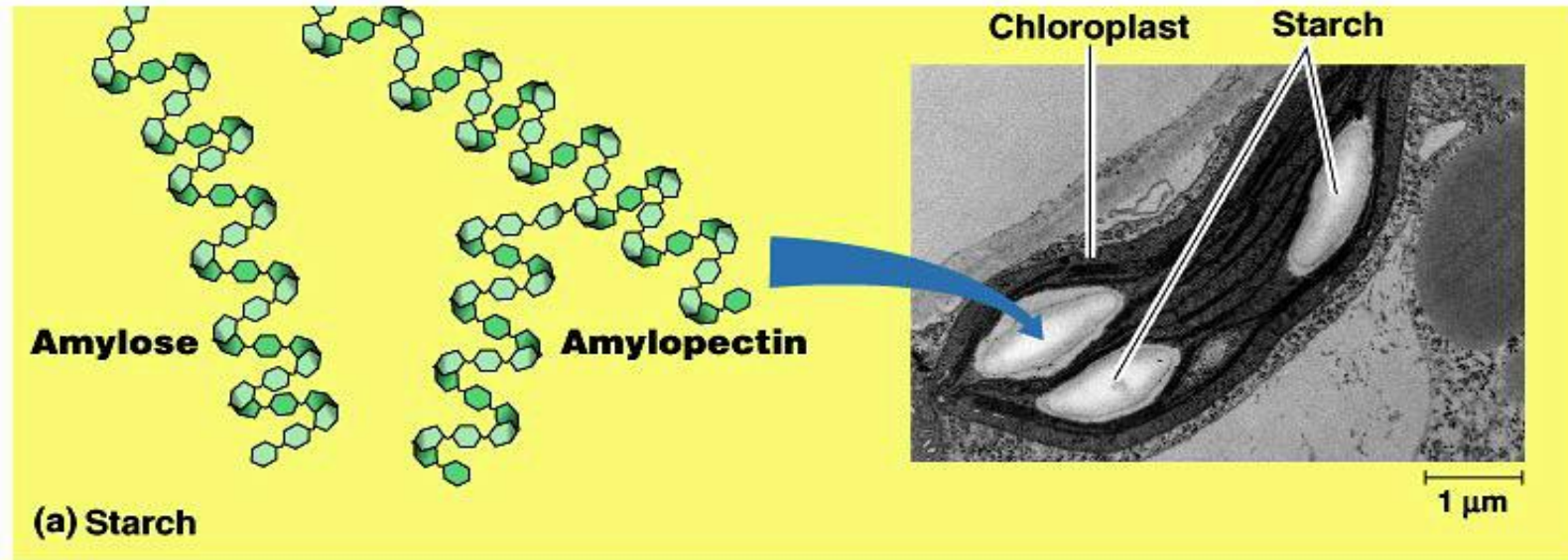
- subunits with a more-or-less rigid structure dictated by covalent bonds form three-dimensional macromolecular structures that are stabilized by weak interactions within or between molecules: hydrogen bonds and hydrophobic and van der Waals interactions, and, for polymers with charged subunits, electrostatic interactions.
- Because polysaccharides have so many hydroxyl groups, hydrogen bonding has an especially important influence on their structure.

# Polysaccharides

- Some homopolysaccharides serve as storage forms of monosaccharides that are used as fuels; starch and glycogen are homopolysaccharides of this type.
- Other homopolysaccharides (cellulose and chitin, for example) serve as structural elements in plant cell walls and animal exoskeletons “provide extracellular support for organisms of all kingdoms”

# Polysaccharides

- The most important storage polysaccharides are starch in plant cells and glycogen in animal cells. Both occur intracellularly as large clusters or granules.
- Starch and glycogen molecules are heavily hydrated, because they have many exposed hydroxyl groups available to hydrogen-bond with water.
- Starch storage is especially abundant in tubers (underground stems), such as potatoes, and in seeds.





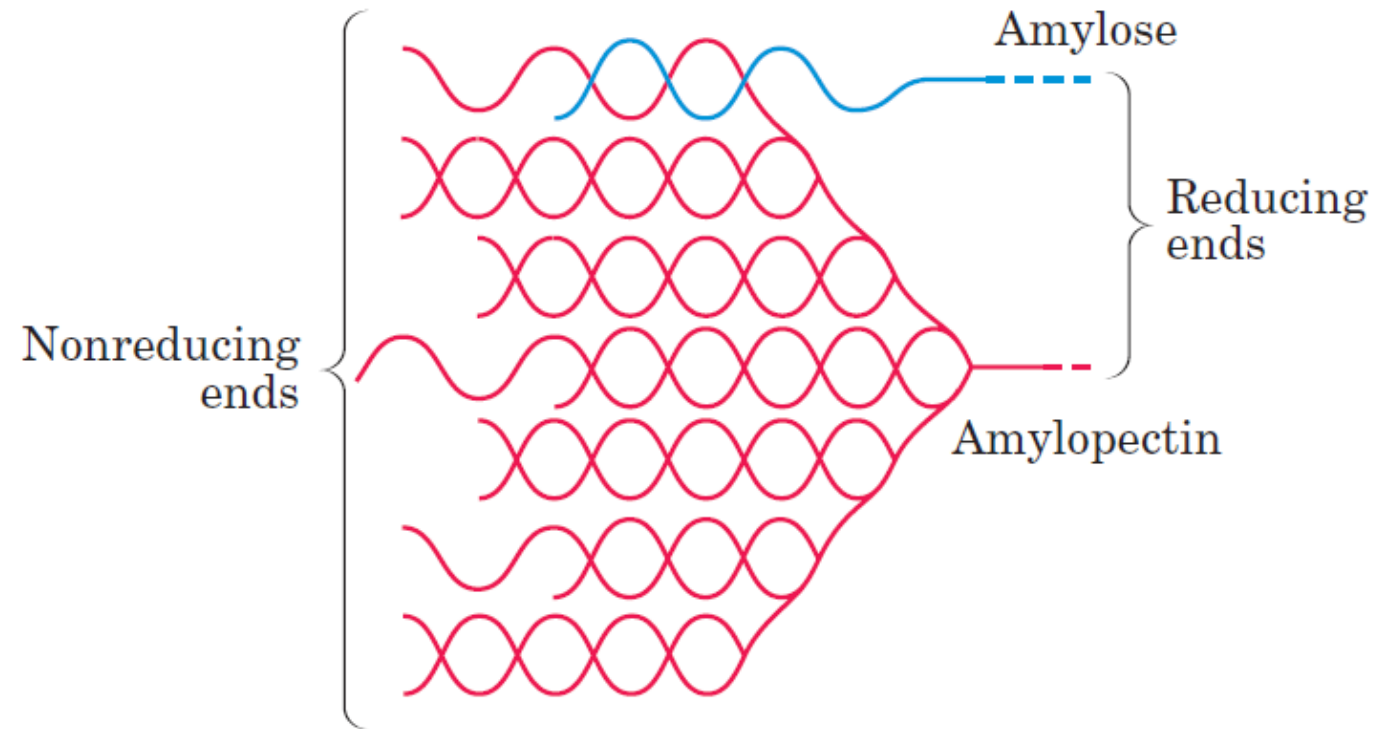
# Starch

- contains two types of glucose polymer, amylose and amylopectin.
- Amylose consists of long, unbranched chains of D-glucose residues connected by ( $\alpha 1 \rightarrow 4$ ) linkages (as in maltose). Such chains vary in molecular weight from a few thousand to more than a million.
- Amylopectin also has a high molecular weight (up to 200 million) and highly branched. The glycosidic linkages joining successive glucose residues in amylopectin chains are ( $\alpha 1 \rightarrow 4$ ); the branch points (occurring every 24 to 30 residues) are ( $\alpha 1 \rightarrow 6$ ) linkages.

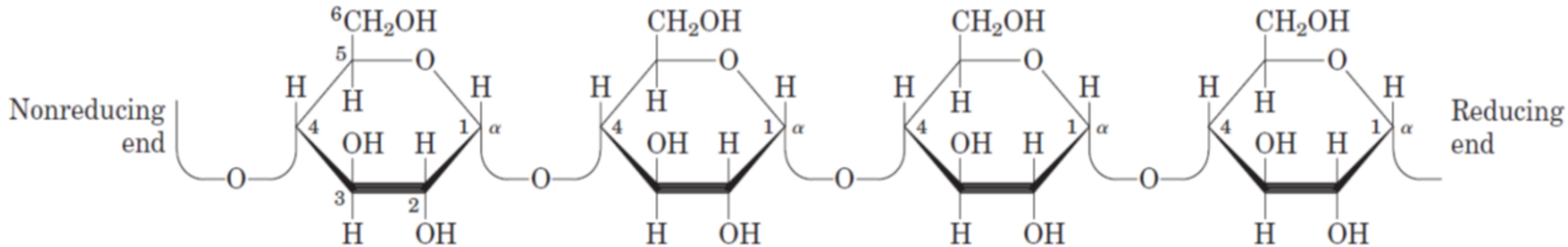
# Starch

- Pure starch is a white, tasteless and odorless
- powder that is insoluble in cold water
- Found in wheat, rice, corn, potatoes
- Gives blue color with iodine

# Starch

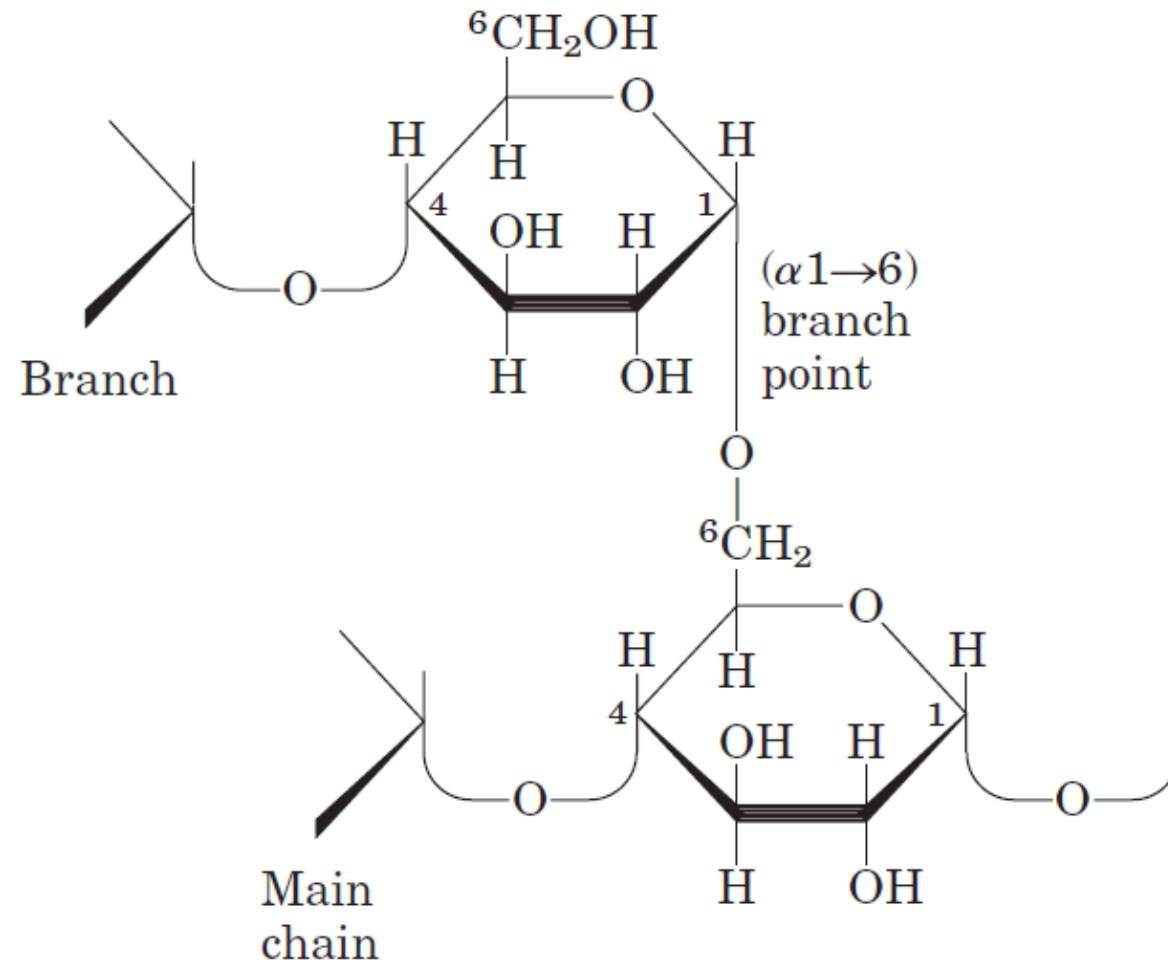


# Starch



(a) Amylose

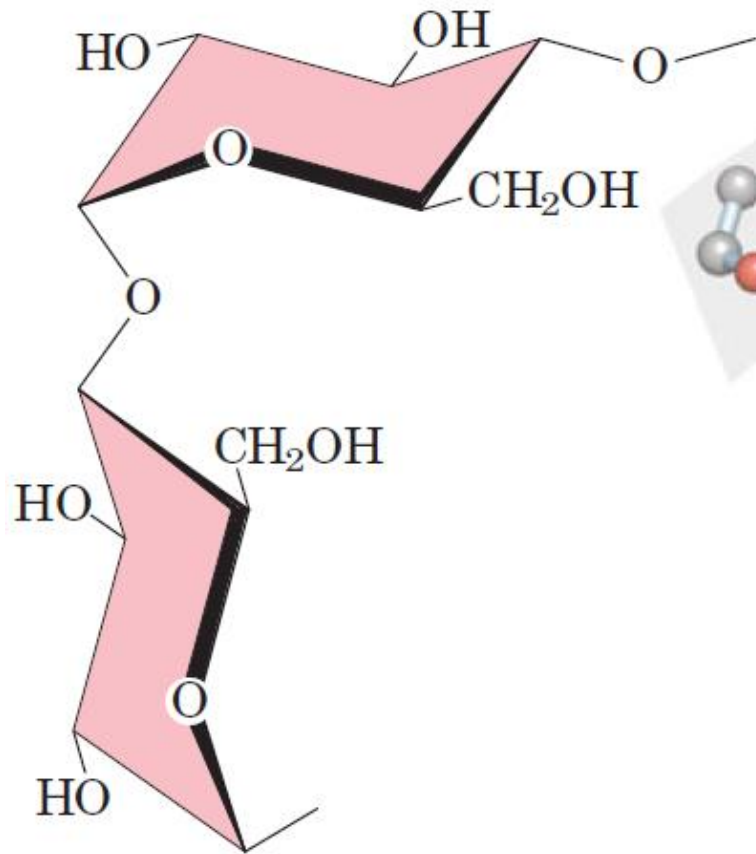
# Starch



# Glycogen

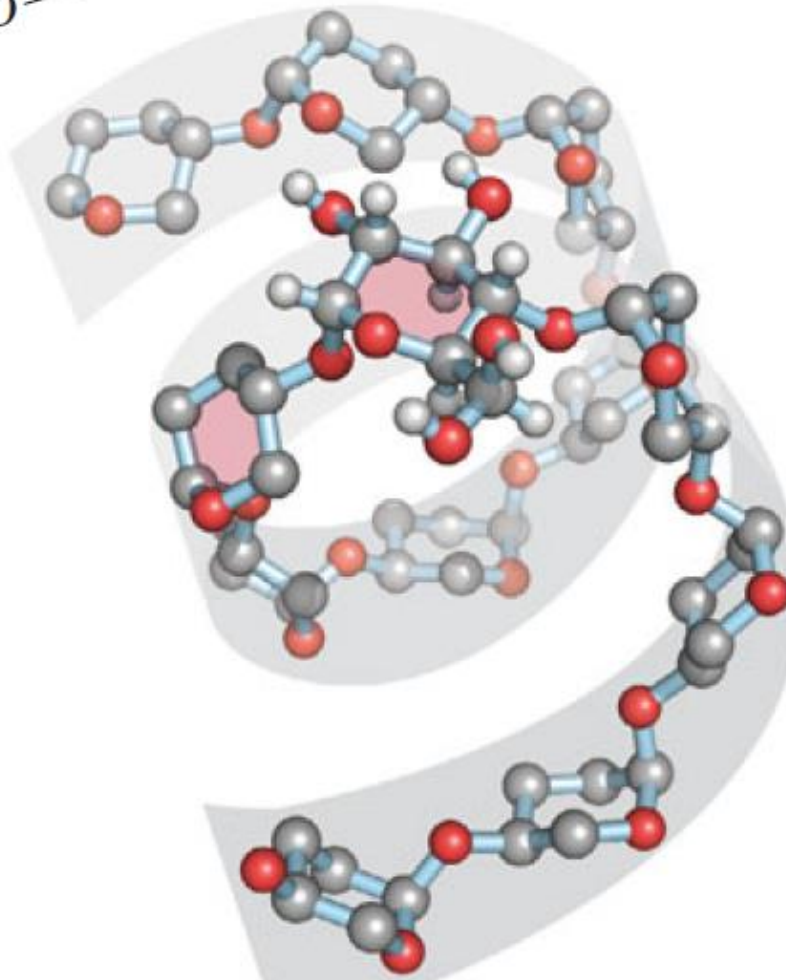
- is the main storage polysaccharide of animal cells.
- Like amylopectin, glycogen is a polymer of ( $\alpha 1 \rightarrow 4$ )-linked subunits of glucose, with ( $\alpha 1 \rightarrow 6$ )-linked branches, but glycogen is more extensively branched (on average, every 8 to 12 residues) and more compact than starch.
- Glycogen is especially abundant in the liver, where it may constitute as much as 7% of the wet weight; it is also present in skeletal muscle

# Starch and Glycogen



( $\alpha 1\rightarrow 4$ )-linked  
D-glucose units

(a)



Dr. Mohamed Osama

(b)

# Dextrans

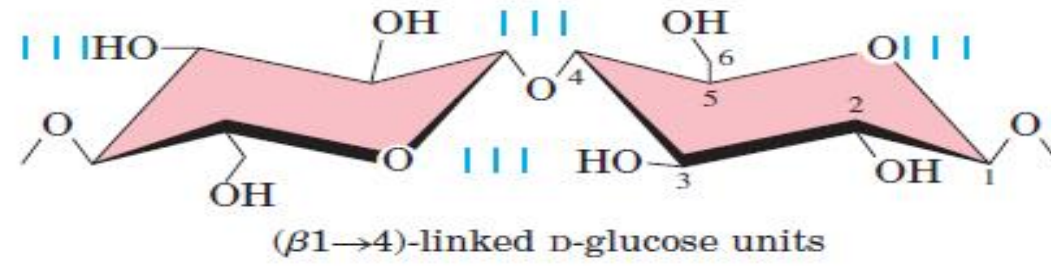
- are bacterial and yeast polysaccharides made up of ( $\alpha 1 \rightarrow 6$ )-linked poly-D-glucose; all have ( $\alpha 1 \rightarrow 3$ ) branches, and some also have ( $\alpha 1 \rightarrow 2$ ) or ( $\alpha 1 \rightarrow 4$ ) branches.
- Dental plaque, formed by bacteria growing on the surface of teeth, is rich in dextrans.
- Synthetic dextrans are used in several commercial products (for example, Sephadex) that serve in the fractionation of proteins by size-exclusion chromatography



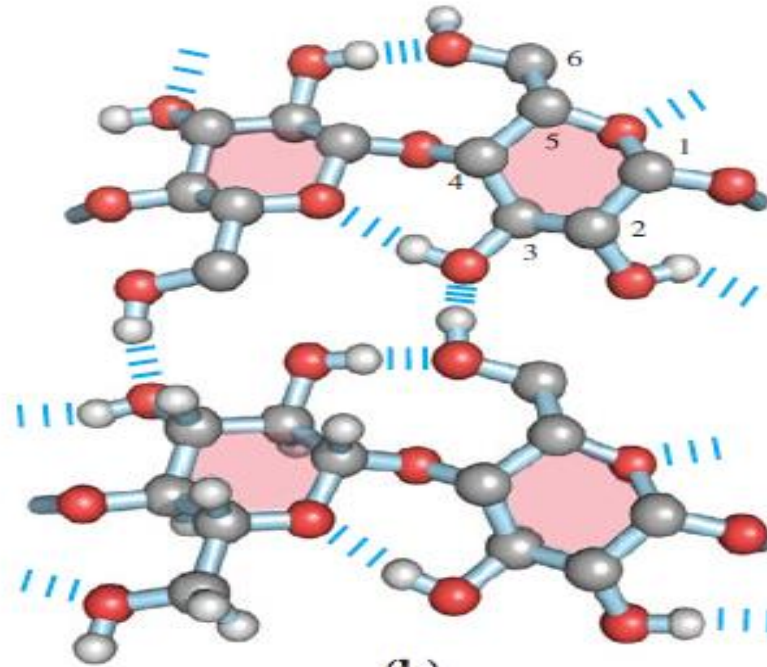
# Cellulose

- a fibrous, tough, water-insoluble substance, is found in the cell walls of plants.
- constitutes much of the mass of wood, and cotton.
- is a linear, unbranched homopolysaccharide, consisting of 10,000 to 15,000 D-glucose units are linked by ( $\beta$ 1 $\rightarrow$ 4) glycosidic bonds.
- But there is a very important difference: in cellulose the glucose residues have the  $\beta$  configuration

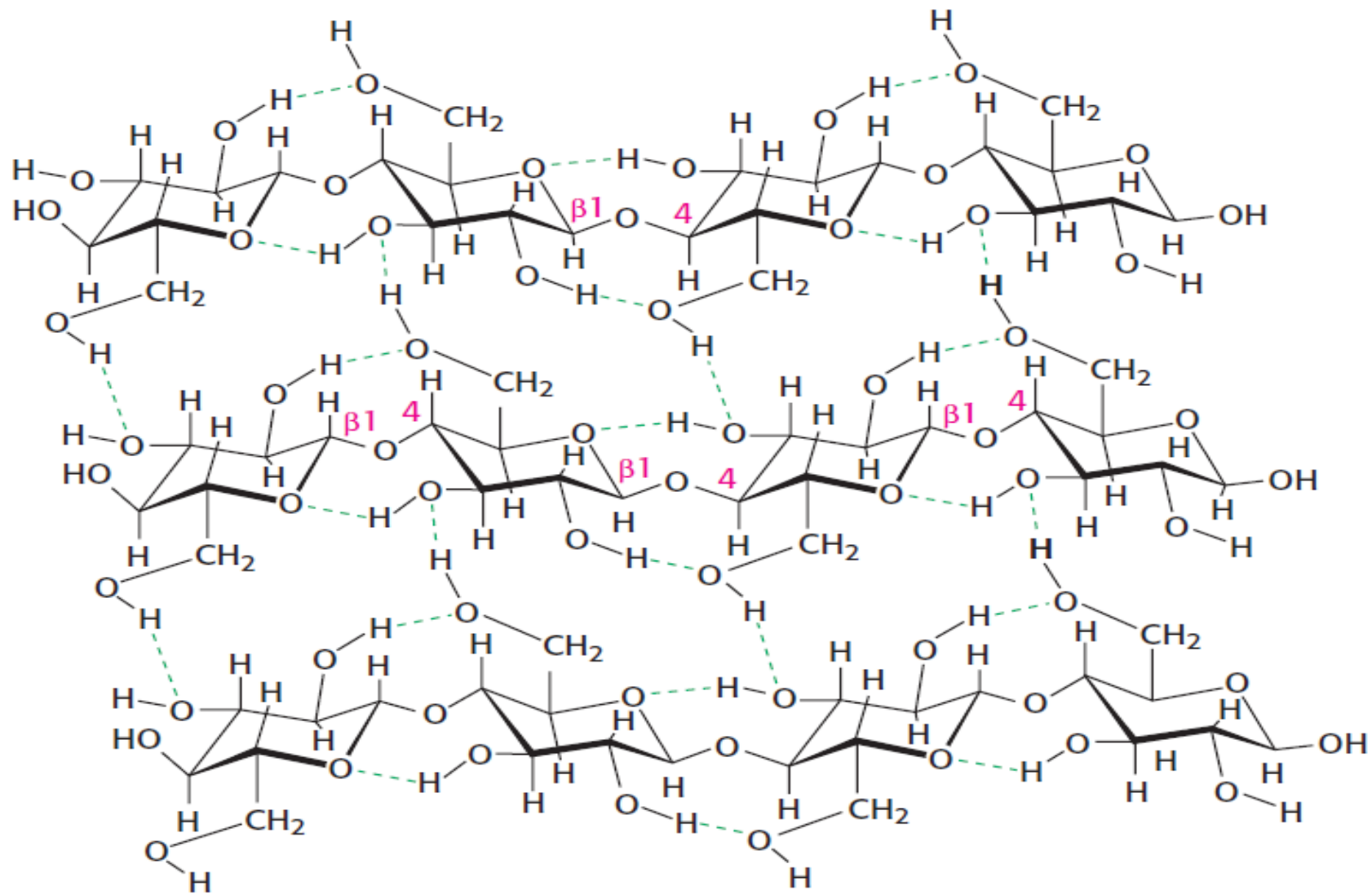
# Cellulose



(a)



(b)



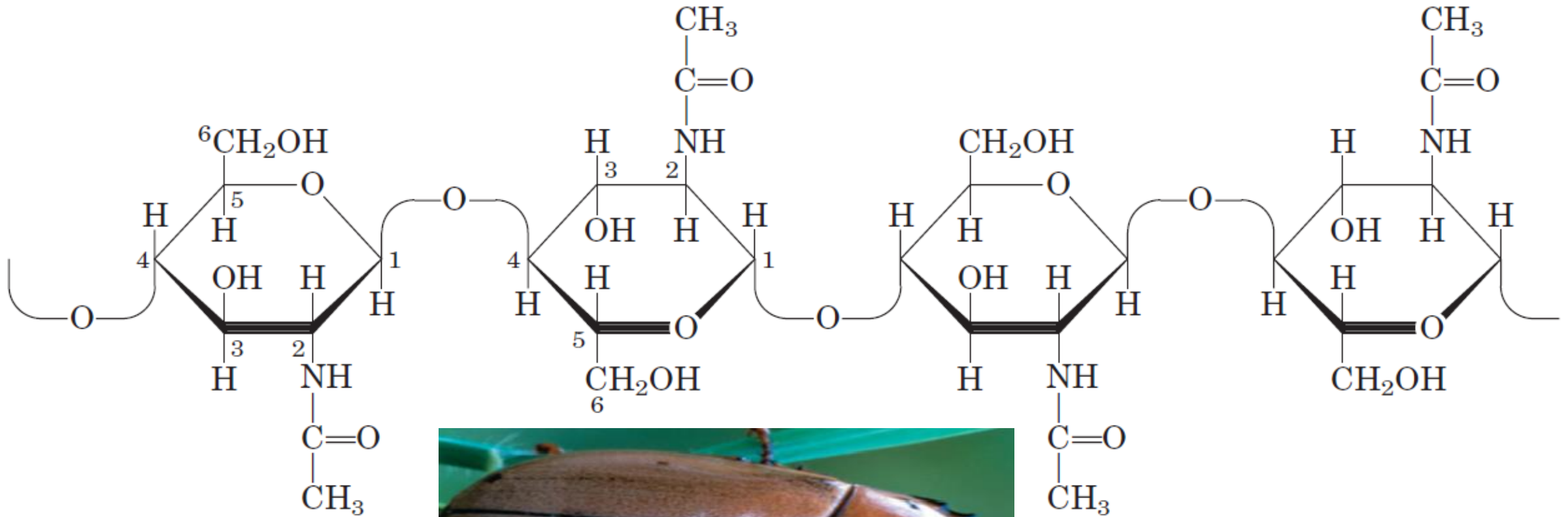
**Cellulose**  
( $\beta$ -1,4 linkages)

- Glycogen and starch ingested in the diet are hydrolyzed by  $\alpha$ -amylases and glycosidases, enzymes in saliva and the intestine that break ( $\alpha 1 \rightarrow 4$ ) glycosidic bonds between glucose units.
- Most animals cannot use cellulose as a fuel source, because they lack an enzyme to hydrolyze the ( $\beta 1 \rightarrow 4$ ) linkages

# Chitin

- is a linear homopolysaccharide composed of *N*-acetylglucosamine residues in ( $\beta$ 1 $\rightarrow$ 4) linkage.
- The only chemical difference from cellulose is the replacement of the hydroxyl group at C-2 with an acetylated amino group.
- Chitin forms extended fibers similar to those of cellulose, and cannot be digested by vertebrates.
- Chitin is the principal component of the hard exoskeletons of nearly a million species of arthropods—insects, lobsters, and crabs

# Chitin



Dr. Mohamed Osama

# Inulin

- It is a polysaccharide of fructose. Found in tubers and roots of some plants, readily soluble in water
- It is used to determine the glomerular filtration rate. But it is not hydrolyzed by intestinal enzymes
- The fructose units in inulin are joined by a  $\beta$  (2 $\rightarrow$ 1) glycosidic bond.
- In general, plant inulin polymers contain between 20 and several thousand fructose units.

# Polysaccharides

- the rigid layer of the bacterial cell envelope (the peptidoglycan) is composed in part of a heteropolysaccharide built from two alternating monosaccharide units.
- In animal tissues, the extracellular space is occupied by several types of heteropolysaccharides, which form a matrix that holds individual cells together and provides protection, shape, and support to cells, tissues, and organs.



- The extracellular space in the tissues of multicellular animals is filled with a gel-like material, the **extracellular matrix (ECM)**, also called ground substance, which holds the cells together and provides a porous pathway for the diffusion of nutrients and oxygen to individual cells.
- The reticular ECM that surrounds fibroblasts and other connective tissue cells is composed of an interlocking meshwork of heteropolysaccharides and fibrous proteins such as fibrillar collagens, elastin, and fibronectin.

- Basement membrane is a specialized ECM that underlies epithelial cells; it consists of specialized collagens, laminin, and heteropolysaccharides.
- These heteropolysaccharides, the **glycosaminoglycans**, are a family of linear polymers composed of repeating disaccharide units. They are unique to animals and bacteria and are not found in plants. One of the two monosaccharides is always either *N*-acetylglucosamine or *N*-acetylgalactosamine; the other is in most cases a uronic acid, usually D-glucuronic or L-iduronic acid.

# Glycosaminoglycans

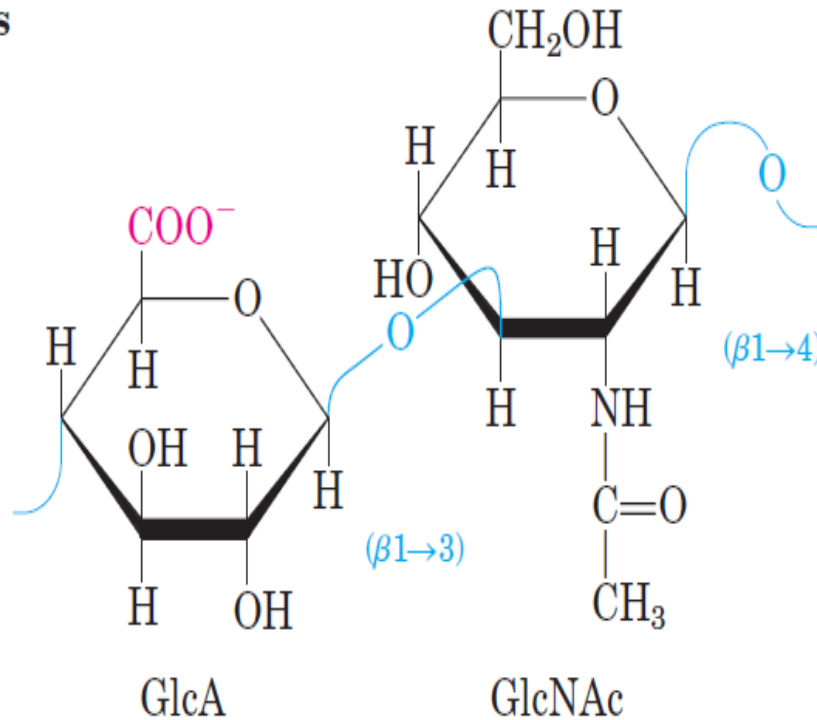
## Glycosaminoglycan

Number of  
disaccharides  
per chain

## Hyaluronan

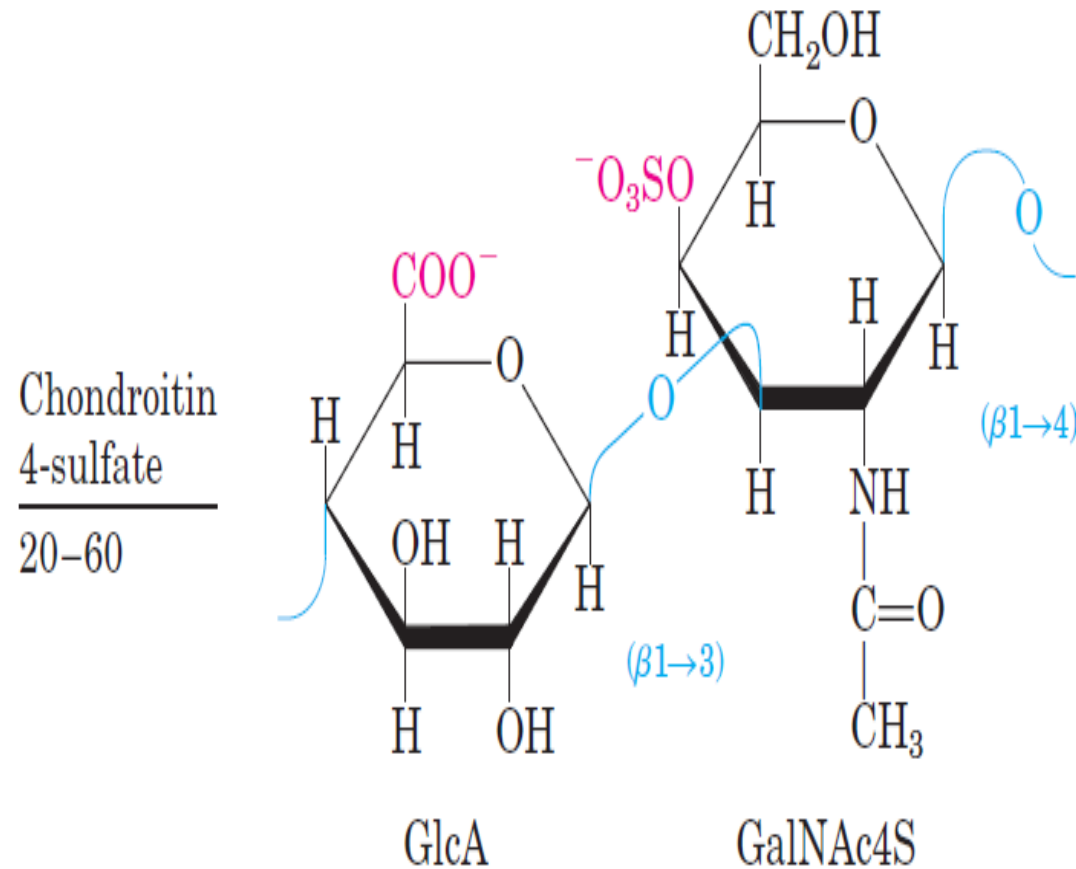
~50,000

## Repeating disaccharide



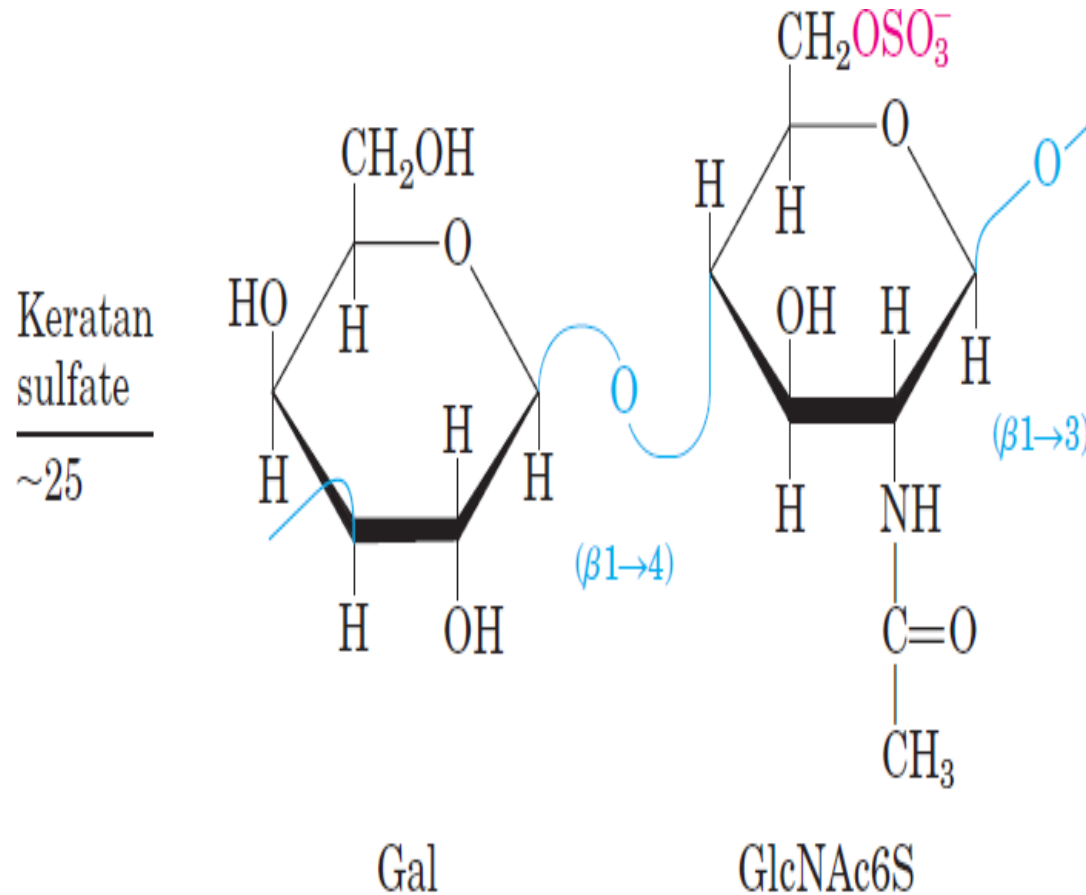
- it forms clear, highly viscous solutions that serve as lubricants in the synovial fluid of joints and give the vitreous humor of the vertebrate eye its jellylike consistency
- (the Greek *hyalos* means “glass”).
- is also a component of the extracellular matrix of cartilage and tendons, to which it contributes tensile strength and elasticity as a result of its strong interactions with other components of the matrix

# Glycosaminoglycans



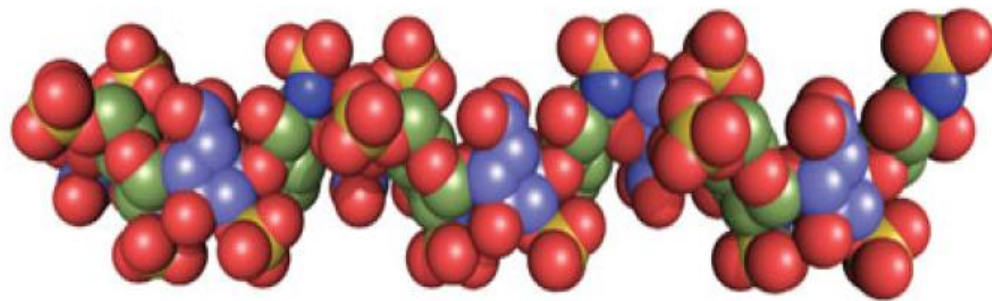
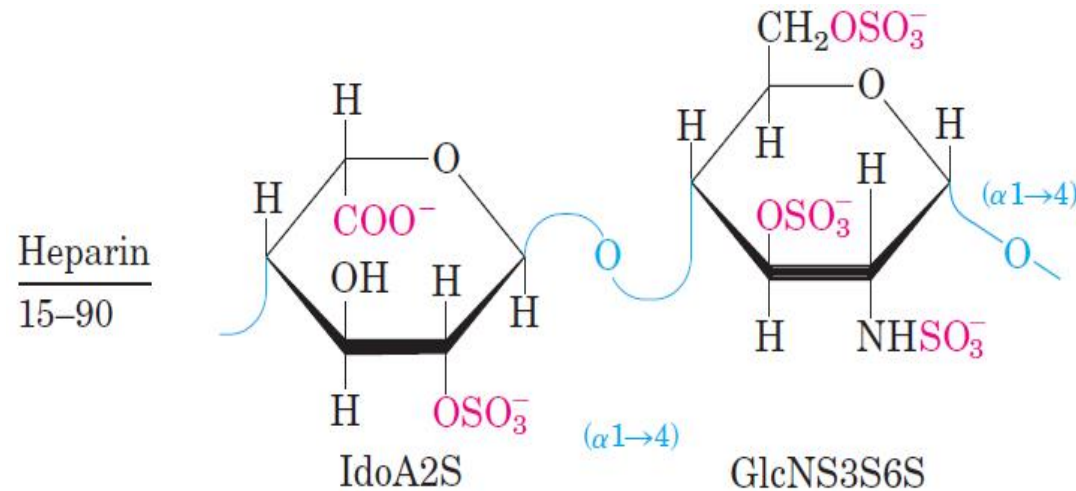
- (Greek *chondros*, “cartilage”)
- contributes to the tensile strength of cartilage, tendons, ligaments, and the walls of the aorta.

# Glycosaminoglycans



- (Greek *keras*, “horn”)
- They are present in cornea, cartilage, bone, and a variety of horny structures formed of dead cells: horn, hair, hoofs, nails, and claws

# Glycosaminoglycans



Heparin segment

- derived mostly from mast cells (a type of leukocyte).
- Heparin is a therapeutic agent used to inhibit coagulation.
- Purified heparin is routinely added to blood samples obtained for clinical analysis, and to blood donated for transfusion.

| TABLE 7–2                                |                                  | Structures and Roles of Some Polysaccharides                                                                     |                                       |                                                                                                                     |
|------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Polymer                                  | Type*                            | Repeating unit†                                                                                                  | Size (number of monosaccharide units) | Roles/significance                                                                                                  |
| Starch<br>Amylose<br>Amylopectin         | Homo-<br>Homo-                   | ( $\alpha$ 1→4)Glc, linear<br>( $\alpha$ 1→4)Glc, with<br>( $\alpha$ 1→6)Glc<br>branches every<br>24–30 residues | 50–5,000<br>Up to $10^6$              | Energy storage: in plants                                                                                           |
| Glycogen                                 | Homo-                            | ( $\alpha$ 1→4)Glc, with<br>( $\alpha$ 1→6)Glc<br>branches every<br>8–12 residues                                | Up to 50,000                          | Energy storage: in bacteria and animal cells                                                                        |
| Cellulose                                | Homo-                            | ( $\beta$ 1→4)Glc                                                                                                | Up to 15,000                          | Structural: in plants, gives rigidity and strength to cell walls                                                    |
| Chitin                                   | Homo-                            | ( $\beta$ 1→4)GlcNAc                                                                                             | Very large                            | Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons                           |
| Dextran                                  | Homo-                            | ( $\alpha$ 1→6)Glc, with<br>( $\alpha$ 1→3) branches                                                             | Wide range                            | Structural: in bacteria, extracellular adhesive                                                                     |
| Peptidoglycan                            | Hetero-;<br>peptides<br>attached | 4)Mur2Ac( $\beta$ 1→4)<br>GlcNAc( $\beta$ 1                                                                      | Very large                            | Structural: in bacteria, gives rigidity and strength to cell envelope                                               |
| Agarose                                  | Hetero-                          | 3)D-Gal( $\beta$ 1→4)3,6-<br>anhydro-L-Gal( $\alpha$ 1                                                           | 1,000                                 | Structural: in algae, cell wall material                                                                            |
| Hyaluronan (a<br>glycosamino-<br>glycan) | Hetero-;<br>acidic               | 4)GlcA( $\beta$ 1→3)<br>GlcNAc( $\beta$ 1                                                                        | Up to 100,000                         | Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints |

# Additional Functions

- In addition to their important roles as stored fuels (starch, glycogen, dextran) and as structural materials (cellulose, chitin, peptidoglycans), polysaccharides and oligosaccharides are information carriers.



# Additional Functions

- Some provide communication between cells and their extracellular surroundings; others label proteins for transport to and localization in specific organelles, or for destruction when the protein is malformed or superfluous; and others serve as recognition sites for extracellular signal molecules (growth factors, for example) or extracellular parasites (bacteria or viruses).

# Additional Functions

- On almost every eukaryotic cell, specific oligosaccharide chains attached to components of the plasma membrane form a carbohydrate layer (the glycocalyx), several nanometers thick, that serves as an information-rich surface that a cell shows to its surroundings.
- These oligosaccharides are central players in cell-cell recognition and adhesion, cell migration during development, blood clotting, the immune response, wound healing, and other cellular processes.

# Additional Functions

- In most of these cases, the informational carbohydrate is covalently joined to a protein or a lipid to form a **glycoconjugate**, which is the biologically active molecule.

# Proteoglycans

- are macromolecules of the cell surface or extracellular matrix in which one or more sulfated glycosaminoglycan chains are joined covalently to a membrane protein or a secreted protein.
- The glycosaminoglycan chain can bind to extracellular proteins through electrostatic interactions with the negatively charged groups on the polysaccharide.
- Proteoglycans are major components of all extracellular matrices

# Glycoproteins

- have one or several oligosaccharides of varying complexity joined covalently to a protein.
- They are usually found on the outer face of the plasma membrane (as part of the glycocalyx), in the extracellular matrix, and in the blood.
- Inside cells they are found in specific organelles such as Golgi complexes, secretory granules, and lysosomes.
- The oligosaccharide portions of glycoproteins are very heterogeneous, forming highly specific sites for recognition and high affinity binding by carbohydrate-binding proteins called lectins.
- Some cytosolic and nuclear proteins can be glycosylated as well.

# Glycolipids

- are membrane sphingolipids in which the hydrophilic head groups are oligosaccharides.
- the oligosaccharides act as specific sites for recognition by lectins.
- The brain and neurons are rich in glycolipids, which help in nerve conduction and myelin formation.
- Glycolipids also play a role in signal transduction in cells.

A still life painting featuring a glass of red wine, a bunch of dark grapes, and several pink roses. The arrangement is set on a light-colored, lace-edged doily against a dark, textured background. The word "Thanks" is written in a large, white, outlined font across the center of the image.

# Thanks